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# Text Encoding in Quantum Natural Language Processing: A Systematic Survey

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**ABSTRACT** Quantum natural language processing (QNLP) applies superposition, entanglement, and interference to the representation and processing of linguistic information. Any QNLP pipeline, whatever its architecture, must first map discrete or distributed text representations onto qubit registers or parameterized-circuit inputs. This encoding step governs model expressivity, qubit resource requirements, and trainability. Yet it is rarely studied as an independent variable, and no unified method exists for analyzing and comparing encodings. This article closes that gap with a systematic literature review of text encoding strategies in QNLP. The review covers 125 primary studies published between 2021 and 2026, under a protocol registered before the search that fixed the search strategy, eligibility criteria, screening procedure, snowball sampling, and data extraction. It introduces a unifying taxonomy for analyzing and comparing encodings. Six encoding families are identified and characterized: angle, amplitude, categorical grammar-based (e.g., DisCoCat), IQP/ZZ feature-map, data re-uploading, and hybrid classical–quantum embedding. Their distribution is analyzed across NLP tasks, simulation and hardware environments, qubit budgets, and performance relative to classical baselines. Three findings stand out: noiseless circuit simulation dominates the literature, real-hardware execution remains rare, and claimed quantum advantage is predominantly conditional or absent. The encoding bottleneck—the dimensionality mismatch between linguistic representations and near-term qubit capacities—is formalized as a basis for resource-efficient QNLP algorithm and hardware design. The corpus and extracted metadata are released publicly to support reproducibility and future meta-analyses.

**INDEX TERMS** Encoding bottleneck, parameterized quantum circuits, quantum advantage, quantum natural language processing, systematic literature review, text encoding.

## I. INTRODUCTION

EVERY quantum natural language processing (QNLP) system begins with a mandatory step: converting text into a quantum state. Before a single gate is applied or a single parameter is trained, a token, sentence, or document must be mapped onto qubits. This mapping—the encoding—determines how many qubits are consumed and how deep the resulting circuit must be. It determines whether gradients can be computed efficiently, and whether any genuinely quantum property, such as entanglement, is brought to bear on the linguistic content. It is, in short, the most consequential design decision in the pipeline.

Despite this centrality, encoding is rarely the subject of dedicated study. Most QNLP papers dispatch it in a paragraph, introduce one strategy without justification, and move on to reporting accuracy numbers. After five years of accelerating publication activity, the result is a fragmented body of work: dozens of encoding families coexist, yet there is no common approach for analyzing and comparing them. Unified taxonomies, systematic comparisons, and a shared understanding of encoding trade-offs are all missing from the literature. Existing QNLP surveys and reviews do not isolate encoding as a variable [1], [2]. The few explicitly comparative studies contrast model families or ansätze rather than encoding schemes [3], [4]. This article fills that gap: it conducts an extensive systematic literature review of the topic and makes text encoding in QNLP the primary subject of study.

### A. SCOPE

This survey reviews primary research that (1) applies a genuine quantum circuit—simulated noiseless, simulated noisy,

or executed on real hardware—to at least one NLP task, and (2) identifies, at least by name or type, the encoding scheme used to map text or text-derived features onto qubits or circuit parameters. The review follows a pre-registered SLR protocol (Section III) adapted from established guidelines for secondary studies in computer science [5].

Papers that use quantum-inspired or quantum-motivated classical algorithms without instantiating an actual parameterized quantum circuit are excluded. Also excluded are survey and review papers, hardware-focused benchmarking studies that treat NLP datasets as a generic payload, and papers with insufficient methodological detail to assess the encoding scheme. The final corpus comprises 125 primary studies.

The scope is limited to the encoding step. Quantum circuit design, ansatz expressibility, and quantum error correction are surveyed only where they interact directly with encoding choices. Classical NLP methods are discussed only as baselines.

### B. KEY CONTRIBUTIONS

This article makes five contributions to the study of text encoding in QNLP:

1) A novel taxonomy of text encoding strategies, organized by the point at which text enters the quantum computation and covering six encoding families with their specialized variants.
2) A pre-registered systematic review, under the Keele and Charters guidelines, of 125 primary studies that apply or propose quantum text encodings across NLP tasks.
3) A formal analysis framework—a classification instrument with fixed dimensions, closed value vocabularies, and explicit decision rules—that operationalizes the taxonomy and extends the process-oriented SLR guidelines with a domain-specific comparison instrument.
4) A quantitative cross-study characterization of the corpus under this framework, spanning encoding-by-model-family, temporal, task-fit, and experimental-rigor analyses, with every count regenerable from the released extraction sheet.
5) A corpus-derived set of seven open challenges, including a formal statement of the encoding bottleneck, an actionable agenda for future work, and the public release of the corpus and extracted metadata.

### C. RELATED WORK

Several prior works survey the QNLP literature, but none treats text encoding as the central variable. Varmantchaonala et al. [6] provide the most comprehensive QNLP survey to date, covering quantum algorithms, circuit architectures, and NLP task categories; encoding appears only as an implementation detail. Guarasci et al. [7] offer an early and influential overview of QNLP challenges and opportunities, from theoretical quantum-advantage arguments to hardware experiments, without isolating encoding as a dimension of comparison. Widdows et al. [8] examine word embeddings and sequential quantum models with more attention to representation trade-offs than any other general survey, but stop short of a systematic comparison. Bouakba and Belhadef [3] experimentally compare compositional DisCoCat models against quantum neural networks; their compositional-versus-flat-vector axis overlaps with the taxonomy proposed here, though their scope is limited to two model families. Huzzat et al. [9] is the most encoding-aware prior work: it addresses the mismatch between text-representation dimensionality and the qubit capacity of NISQ classifiers, and treats classical compression as a prerequisite for stable training. It remains, however, a conceptual primer for a single model class rather than a systematic review.

Beyond QNLP itself, the quantum engineering literature has an established tradition of comprehensive subfield surveys, covering, for instance, quantum circuit compilation for NISQ devices [10], quantum computing for finance [11], and entanglement routing in quantum networks [12], the last of which likewise organizes its field through an explicit taxonomy of approaches. None of these addresses text encoding in QNLP, which is the gap the present survey fills within that tradition.

This survey complements these works by treating encoding as the primary unit of analysis across 125 primary studies. Encoding family, qubit count, NLP task, execution environment, and quantum advantage claim were extracted for each study, yielding the first quantitative cross-study characterization of how text is encoded and at what resource cost.

### D. STRUCTURE

The rest of the article is organized as follows. Section II provides the background on quantum computing, compositional distributional semantics, and the role of encoding in QNLP. Section III describes the systematic review protocol—search strategy, eligibility criteria, screening, and snowball sampling—and presents the encoding comparison framework used throughout the analysis. Section IV introduces the taxonomy of text encoding strategies and characterizes each family. Section V analyzes how each encoding family is used across model families and ansätze. Section VI traces the temporal evolution of encoding and model choices. Section VII examines the fit between encoding, model family, and reported quantum advantage at the task level. Section VIII assesses experimental rigor across simulation regimes, hardware use, and reproducibility. Section IX synthesizes the trends, states the encoding bottleneck formally, and enumerates the open challenges and the future-work agenda. Section X concludes. Appendix A lists the complete per-study classification of the corpus.

## II. BACKGROUND

### A. QUANTUM COMPUTING PRIMITIVES

A quantum bit (qubit) occupies a superposition of the basis states $|0\rangle$ and $|1\rangle$:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1, \tag{1}$$

where $\alpha, \beta \in \mathbb{C}$ are probability amplitudes. A register of $n$ qubits occupies a $2^n$-dimensional Hilbert space $\mathcal{H} = \mathbb{C}^{2^n}$; its state is a unit vector in that space. This exponential dimension is the property every amplitude-style encoding exploits. Entanglement introduces correlations with no classical counterpart, and measurement collapses the state to a classical bitstring, destroying superposition. Whether superposition, entanglement, or interference confers a practical advantage on NLP tasks is the field's central open question and one of the dimensions analyzed in this survey.

### B. PARAMETERIZED QUANTUM CIRCUITS

The dominant computational model in near-term quantum machine learning is the parameterized quantum circuit (PQC), also called a variational quantum circuit [13]. A PQC applies a sequence of unitary gates to an initial state $|\psi_0\rangle$:

$$U(\boldsymbol{\theta}) = U_L(\boldsymbol{\theta}_L) \cdots U_1(\boldsymbol{\theta}_1), \tag{2}$$

where $\boldsymbol{\theta} \in \mathbb{R}^p$ are trainable parameters. Model outputs are expectation values $\langle O \rangle = \langle \psi_0 | U^\dagger(\boldsymbol{\theta})\, O\, U(\boldsymbol{\theta}) | \psi_0 \rangle$, and gradients follow analytically from the parameter-shift rule [14], [15]. PQCs underlie five of the six encoding families surveyed here; the exception is grammatical encoding, which derives circuit structure from sentence syntax rather than from a fixed ansatz.

PQCs operate under the constraints of noisy intermediate-scale quantum (NISQ) hardware [16]: shallow circuits, limited qubit counts, and significant gate error rates. These constraints bear directly on the encoding step. Every qubit or gate layer spent loading a text representation is a resource unavailable to the learning task.

### C. TEXT ENCODING

Text encoding is the procedure that maps a text input $x$—a token, sentence, or document—to a quantum state $|\phi(x)\rangle \in \mathcal{H}$ or to a parameter vector $\boldsymbol{\theta}(x)$ that initializes a PQC. It is the mandatory interface between classical language data and the quantum circuit. The input $x$ may be a raw token sequence, a classical embedding vector from a pre-trained model, or a parse tree carrying grammatical structure. Which representation is used, and how it is loaded onto qubits, defines the encoding family.

Encoding choices have cascading consequences. The required qubit count, the depth of the state-preparation circuit, the presence of linguistically meaningful entanglement, and the trainability of the resulting model all depend on this single decision. Most QNLP papers nonetheless treat encoding as a fixed implementation detail rather than as a variable to be studied.

### D. COMPOSITIONAL DISTRIBUTIONAL SEMANTICS AND DISCOCAT

One of the two main traditions in QNLP originates in categorical linguistics. Coecke, Sadrzadeh, and Clark [17] showed that the compact closed categorical structure of pregroup grammars also governs quantum processes in Hilbert space. In a pregroup grammar, each word carries a grammatical type, and a sentence is parsed by reducing compound types through tensor contractions. These reductions obey the same diagrammatic calculus as quantum processes, so a parse tree translates directly into a quantum circuit: words become parameterized quantum states, and grammatical reductions become cups in a string diagram.

This is the DisCoCat (Distributional Compositional Categorical) model, operationalized as an end-to-end pipeline by [18] and executed on near-term hardware in [19], [20] via the lambeq toolkit. The circuit structure is not fixed in advance: the parse tree determines it, so syntactically distinct sentences produce structurally distinct circuits. This property sets the grammatical encoding family apart from all other families, which treat the sentence as a flat feature vector.

### E. QUANTUM MACHINE LEARNING FOR TEXT

The second tradition approaches NLP as a classification problem. It applies quantum feature maps [21] and variational circuits [13] to text-derived feature vectors: classical embeddings (word2vec, GloVe, or transformer outputs) are compressed to the available qubit count, loaded into a PQC via rotation gates or amplitude encoding, and trained end-to-end on labeled examples. Because this approach assumes nothing about grammatical structure, it applies to arbitrary text without a parser and scales more easily to realistic dataset sizes. Its logical endpoint is the hybrid quantum–classical architecture, in which a large classical language model handles high-dimensional representation and a small quantum circuit acts as a downstream head; such architectures have grown in prevalence since 2023.

The two traditions differ not only in how they encode text but in what they claim the quantum component contributes. Compositional models argue for structural alignment between grammar and quantum computation; QML models argue for expressivity or kernel separability.

## III. REVIEW METHODOLOGY

This systematic literature review followed the guidelines for secondary studies in software engineering established by Keele and Charters [5]. The protocol was defined before the search was executed. It fixed the search strategy, eligibility criteria, screening procedure, snowball sampling, and data extraction process.

### A. SEARCH STRATEGY

Two complementary search queries were executed on two electronic databases, Scopus and IEEE Xplore, to cover both the quantum machine learning and the compositional QNLP literature. The first query targeted the intersection of quantum

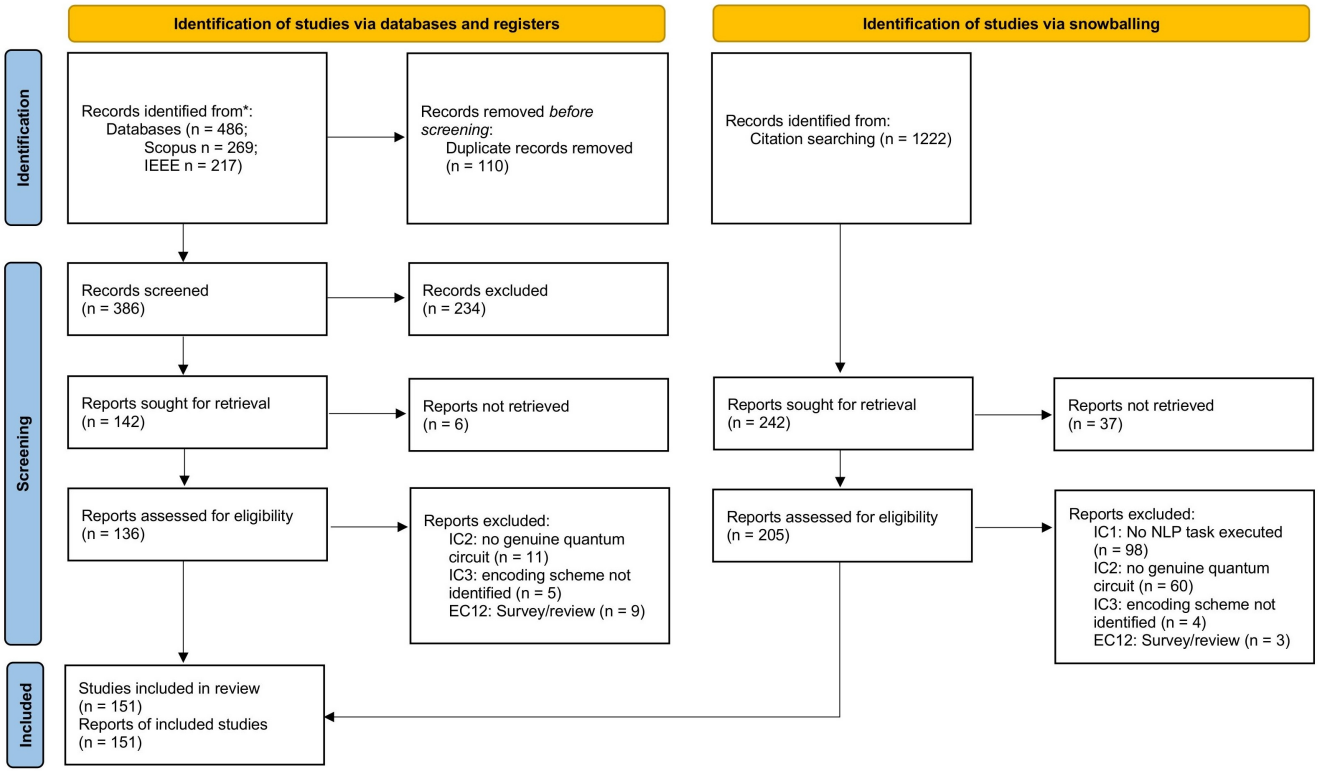



**FIGURE 1.** PRISMA flow diagram. Final corpus: $n = 125$ (95 from database search, 30 from snowball sampling).

computing or machine learning with NLP and encoding-related terms. The second targeted QNLP-specific terminology: the DisCoCat and DisCoCirc frameworks, established pipelines that implement them such as the lambeq toolkit, and categorical linguistic concepts. The searches were restricted to English-language journal articles and conference papers published between 2021 and 2026 (ends included), excluding editorials, book chapters, and erratum notices. They returned 269 results from Scopus and 217 from IEEE Xplore, a combined total of 486 records. Removing 110 duplicates left 376 unique records.

### B. ELIGIBILITY CRITERIA

A study was selected if and only if it met all six inclusion criteria below.

- *IC1: NLP task grounding.* The study applies or evaluates the proposed method on at least one natural language processing task, such as classification, similarity, sequence labeling, question answering, or generation.
- *IC2: Genuine quantum computation.* The study implements and evaluates the method in at least a noiseless quantum circuit simulation (exact statevector simulation or equivalent) or on real quantum hardware; merely quantum-inspired classical algorithms, such as tensor-network models or classical approximations of quantum kernels, do not satisfy this criterion.
- *IC3: Encoding scheme identified.* The study names or identifies at least the type of encoding used to map text or text-derived features onto qubits, quantum states, or circuit parameters; a named or typed mention is sufficient, and circuit diagrams or full mathematical derivations are not required.
- *IC4: Peer-reviewed or preprint venue.* The study appeared in a peer-reviewed journal, conference, or workshop, or as a preprint with sufficient methodological detail to assess IC1–IC3.
- *IC5: Language and access.* The study is available in full text, in English, and accessible for review.
- *IC6: Time window.* The study was published between 2021 and 2026 inclusive.

Even if a study satisfied all six inclusion criteria, it was excluded if any of the following applied:

- *EC1: No NLP application.* The study applies quantum machine learning exclusively to non-text domains and offers no transferable text-encoding discussion.
- *EC2: Classical-only encoding.* The study uses encoding purely in the classical sense (tokenization, positional encoding, one-hot) with no quantum circuit involved.
- *EC3: Insufficient detail.* The study is an abstract, extended abstract, or short paper that names an encoding but provides no formal description or reproducible detail.
- *EC4: Duplicate or superseded.* The study is a confer-

ence version superseded by an extended journal version; only the more complete version was retained.
- *EC5: Pure hardware focus.* The study addresses hardware benchmarking alone and uses an NLP dataset only as a generic payload, with no discussion of linguistic encoding properties.
- *EC6: Quantum-inspired, classical surrogate, or unverifiable simulation.* The study uses quantum-derived mathematical formalism without instantiating an actual parameterized quantum circuit, reports only classical surrogates of quantum behavior, or claims quantum simulation without evidence of framework or circuit use.
- *EC7: Survey or review paper.* The study is primarily a secondary study rather than an original research contribution; such papers were retained as reference material and are discussed in Section I-C.

### C. SCREENING AND SELECTION

Screening proceeded in two stages. In the first stage, all 376 records were assessed on title and abstract alone for broad topical relevance to quantum computing applied to text. No IC/EC criteria were applied at this stage; its only purpose was to remove records clearly outside the scope of the review. This eliminated 234 records and left 142 reports sought for retrieval, of which 6 could not be retrieved, leaving 136 reports assessed for eligibility.

In the second stage, the full text of each of the 136 reports was assessed against the complete set of inclusion and exclusion criteria of Section III-B. This yielded 95 papers for the primary corpus. The remaining 41 retrieved reports were excluded or retained as surveys, mainly under IC2 (no genuine quantum circuit; $n = 15$), IC3 (encoding scheme not identified; $n = 6$), EC6 (quantum-inspired, classical surrogate, or unverifiable simulation; $n = 3$), and EC7 (survey or review; $n = 11$); six records were excluded under other criteria.

### D. SNOWBALL SAMPLING

To reduce the risk of missing relevant work not indexed by the two databases, two rounds of backward and forward snowball sampling [22] were applied to the surveys excluded under EC7. Reference lists and citing papers yielded 1222 unique candidate records. Title-and-abstract screening for broad topical relevance reduced this set to 242 reports sought for retrieval, of which 37 could not be retrieved, leaving 205 reports assessed for eligibility. Full-text assessment against the complete IC/EC protocol then added 30 papers to the corpus. The remaining 175 retrieved reports were excluded or retained as surveys, mainly under IC1 (no NLP task; $n = 74$), IC2 (no genuine quantum circuit; $n = 52$), IC6 (outside the time window; $n = 20$), IC3 (encoding scheme not identified; $n = 4$), and EC7 (survey or review; $n = 22$); three records were excluded under other criteria.

### E. FINAL CORPUS

The final primary corpus comprises 125 papers: 95 from database search and 30 from snowball sampling. Fig. 1 summarizes the selection process following the PRISMA flow diagram convention.

### F. ANALYSIS FRAMEWORK: THE ENCODING COMPARISON INSTRUMENT

The guidelines of Keele and Charters [5] prescribe the process of a secondary study. The preceding subsections instantiate each of its stages: planning and protocol definition (registered before the search), search strategy, study selection, and, through the extraction schema described below, data extraction and synthesis. What the guidelines deliberately do not prescribe is a domain-specific comparison instrument—a fixed scheme by which the primary studies of a particular field are classified and made commensurable. For a review whose object is the encoding step itself, such an instrument is the methodological core, and its design is a contribution of this survey in its own right. The remainder of the article refers to it as the encoding comparison framework. Every classification, cross-tabulation, and trend reported in Sections IV–IX is an application of this framework, not an ad hoc reading of the corpus.

The framework classifies every primary study along a fixed set of dimensions, each with a closed value vocabulary (Table 1). The two primary dimensions are the encoding family, defined by the point at which text or text-derived features enter the quantum computation (Section IV), and the model family, defined by the architecture that consumes the encoded state. Secondary dimensions capture the resources and evidence attached to that choice: qubit count and whether qubit scaling was tested, execution regime, presence of a classical baseline, the paper's own quantum-advantage claim, and three reporting dimensions (encoding-bottleneck awareness, barren-plateau treatment, scalability discussion). Each dimension is operationalized as one or more columns of the public extraction sheet released with this survey. The framework and the dataset are therefore two views of the same instrument.

Assignment follows explicit decision rules. A study is classified by its dominant encoding interface: the one that carries the text representation into the circuit whose output is evaluated. Auxiliary encodings used only in ablations or baselines do not change the class; a study whose main pipeline is amplitude-encoded but that reports an angle-encoded ablation is classified as amplitude. Where a pipeline stacks a classical encoder in front of a quantum layer, the study is classified as `hybrid_embed` only when the classical stage is co-trained or purpose-built as part of the contribution; a frozen pretrained embedding feeding a standard rotation layer remains `angle`. The `hybrid_nn` model family covers any architecture in which a classical deep network (transformer, recurrent, or convolutional—not only LLM-based extractors) performs the main representational work and the quantum circuit is a submodule. A bare variational

classifier over precomputed features is `vqc`, even when those features come from a deep model. Compositional (DisCoCat-style) studies are identified by circuit topology derived from a grammatical parse, regardless of toolkit. These disambiguation rules were fixed during protocol definition and applied uniformly during extraction.

Extraction and classification were performed by one author and reviewed by a second; disagreements were resolved by discussion against the decision rules above, and each row of the extraction sheet carries an explicit confidence grade (high/medium/low) reflecting how completely the primary study reports the extracted fields. Formal inter-rater reliability (e.g., Cohen's $\kappa$) was not computed, which we state as a limitation of the extraction process rather than estimate post hoc.

## IV. TAXONOMY OF TEXT ENCODING STRATEGIES

The encoding problem in QNLP is not a single operation. Depending on the study, text can enter the quantum computation as rotation parameters, as amplitudes of a prepared state, as a grammar-derived circuit, as a structured feature map, or through a more specialized interface such as repeated data injection, a hybrid neural representation, or an observable. These mechanisms are often discussed under the same term, "encoding," although they make different assumptions about where linguistic information is represented and which quantum resource is being used.

This section shows how these mechanisms are organized into the proposed taxonomy as shown in Table 2, applying the encoding comparison framework of Section III-F: every study discussed below is assigned to exactly one encoding family by the framework's dominant-interface decision rule. The classification is based on the point at which text or text-derived features enter the quantum computation. We therefore do not classify a study by the optimizer, measurement rule, downstream classifier, or ansatz alone. These elements are treated as design choices that can be combined with an encoding category. This distinction is important because an applied QNLP pipeline is usually a combination of several decisions: a classical text representation, a quantum encoding interface, a trainable or fixed circuit, and a measurement or kernel rule.

The final analyzed corpus contains 125 primary studies. Four major encoding families are observed: angle encoding ($n = 51$), amplitude encoding ($n = 28$), syntactic/DisCoCat encoding ($n = 29$), and structured feature-map encoding ($n = 9$). A smaller residual group ($n = 8$) contains specialized interfaces: data re-uploading ($n = 4$), hybrid classical–quantum embedding ($n = 3$), and Hamiltonian/observable encoding ($n = 1$). To use the taxonomy, one may first identify whether the text representation is loaded as a flat vector, imposed through linguistic structure, or used to define a kernel geometry. The next design question is whether the encoding is intended to minimize qubit count, preserve linguistic composition, define a kernel geometry, or improve trainable circuit expressivity. Not all combinations are equally practical on near-term devices. The main tradeoff is between width, depth, state-preparation cost, and the amount of linguistic structure preserved by the quantum circuit [13], [16], [23].

### A. ANGLE ENCODING

Angle encoding is the dominant category in the corpus. In this scheme, a classical vector $\mathbf{x} = (x_1, \ldots, x_d)$ is mapped to rotation parameters,

$$|\phi(\mathbf{x})\rangle = \prod_{j=1}^{d'} R_{\alpha_j}(s_j(\mathbf{x}))|0\rangle^{\otimes n}, \tag{3}$$

where $R_{\alpha_j}$ is a rotation around the $x$, $y$, or $z$ axis and $s_j(\cdot)$ denotes scaling, projection, or feature selection. The input is rarely raw text. It is usually a TF-IDF vector, a word or sentence embedding, a transformer representation, or a compact feature vector produced by a preceding classical model.

The prevalence of angle encoding follows from its implementation simplicity. Rotations fit naturally into parameterized quantum circuits and support the standard gradient-based training workflow of variational algorithms, and they avoid arbitrary-state preparation, which makes them convenient for small simulated or hardware-oriented experiments [14], [15]. What makes the family a single encoding category is this shared rotation mechanism; within it, the corpus studies differ along two encoding-level axes: the rotation construction itself and the classical source that supplies the angles.

The first axis is the rotation construction. The most common pattern assigns one feature to one single-axis rotation, typically an $R_Y$ rotation as in the default angle-embedding routine [23]–[25], though $R_X$ is also used [26]–[28] and some studies apply combined $R_Y$ and $R_Z$ rotations per qubit [29]. A denser variant stores more than one feature on each qubit through successive rotations about different axes, trading a shorter register for greater loading depth [30].

The second axis is the classical source of the angles, and it is where the studies vary most, because the rotation parameters are numbers rather than text and so the encoding's content is fixed by the front-end that produces them. Some studies load sparse lexical statistics such as TF-IDF, bag-of-words, or $n$-gram vectors [31]–[33]; others load static word embeddings [25], [34], [35]; and a large, growing group compresses contextual transformer embeddings, for instance reducing a 768-dimensional BERT or DistilBERT vector to a handful of features before rotation [36]–[39]. A further group derives the angles from deeper classical feature extractors or from handcrafted linguistic features [40]–[42]. In each case a dimensionality-reduction step, whether PCA, a linear projection, or an autoencoder, sits between the text representation and the rotations and is effectively part of the encoding, since it determines what survives into the quantum circuit.

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**TABLE 1.** The encoding comparison framework: classification dimensions, admissible values, and the decision rule that assigns a primary study to a value. Each dimension is realized as one or more columns of the released extraction sheet.

| Dimension | Admissible values | Decision rule |
|---|---|---|
| Encoding family | angle; amplitude; syntactic (DisCoCat); structured feature map; data re-uploading; hybrid embedding; Hamiltonian/observable | Dominant interface by which text-derived data enters the evaluated circuit; ablation-only encodings ignored |
| Model family | VQC; compositional; hybrid neural; quantum transformer; quantum kernel; recurrent neural | Architecture consuming the encoded state; classical-deep-network host with quantum submodule ⇒ hybrid neural; parse-derived topology ⇒ compositional |
| Qubit resources | qubit count (integer or NR); scaling tested (yes/no) | Largest register of the evaluated text circuit as reported; scaling = yes only if more than one register width is evaluated |
| Execution regime | noiseless simulation; noisy simulation; mixed; real hardware | Highest-fidelity regime in which the main result is obtained; mixed = simulation plus any hardware validation run |
| Baseline & advantage | classical baseline (yes/no); advantage claim (yes/conditional/no) | Advantage claim as stated by the paper's own authors against their own baseline; not re-derived |
| Reporting dimensions | bottleneck (explicit/implicit/absent); barren plateaus (discussed/mentioned/absent); scalability (yes/no) | Explicit = named and analyzed; implicit = the constraint shapes design without being named; mentioned = term appears without engagement |

**TABLE 2.** Taxonomy of text encoding strategies in the corrected primary corpus.

| Encoding family | Corpus count | Core mechanism | Typical model families |
|---|---|---|---|
| Angle encoding | 51 | Text features are mapped to single-qubit rotation parameters, typically through $R_x$, $R_y$, or $R_z$ gates. | VQC, hybrid neural, recurrent neural, quantum transformer |
| Amplitude encoding | 28 | A normalized feature vector is loaded into the amplitudes of an $n$-qubit state in a $2^n$-dimensional Hilbert space. | VQC, quantum kernel, quantum transformer, hybrid neural |
| Syntactic/DisCoCat encoding | 29 | Sentence grammar determines a string diagram or circuit; word meanings are attached to grammatical types and composed by reductions. | Compositional QNLP |
| Structured feature maps | 9 | Classical text features parameterize structured circuits such as IQP, ZZFeatureMap, ZFeatureMap, PauliFeatureMap, or Pauli-Y variants. | Quantum kernel, VQC, hybrid neural, quantum transformer |
| Data re-uploading | 4 | Input features are injected repeatedly across circuit layers, alternating data-dependent rotations with trainable transformations. | VQC, recurrent neural |
| Hybrid embedding | 3 | A classical encoder first produces a compact text representation that is then passed to a quantum layer. | Hybrid neural, VQC |
| Hamiltonian/observable encoding | 1 | Text-derived information is represented through Pauli-string Hamiltonians or observables rather than only through prepared quantum states. | VQC |

The limitation is inherent in this design. Encoding cost grows with the number of retained features, so a model must spend more rotations, more qubits, or more layers as it keeps more of the input. Angle encoding therefore pushes the hard part of the problem onto classical dimensionality reduction, which makes it practical for NISQ-scale prototypes but also means the quantum circuit often operates on a heavily compressed view of the linguistic input.

### B. AMPLITUDE ENCODING

Amplitude encoding maps a normalized vector $\mathbf{x} \in \mathbb{R}^{2^n}$ or $\mathbb{C}^{2^n}$ to a quantum state,

$$|\phi(\mathbf{x})\rangle = \sum_{i=0}^{2^n-1} x_i |i\rangle, \quad \sum_i |x_i|^2 = 1. \tag{4}$$

For NLP, the appeal is immediate. A high-dimensional sentence embedding can, in principle, be represented with logarithmically many qubits: a 768-dimensional transformer vector, for example, can be padded and normalized into a ten-qubit state. This density is why amplitude encoding is most often paired with dense contextual representations, above all BERT, ELECTRA, and other transformer embeddings, and to a lesser extent with static word embeddings or TF-IDF and bag-of-words vectors [43]–[46].

Because the register holds exactly $2^n$ amplitudes, the encoding is defined as much by how the source vector is fitted to that dimension as by the loading step itself. When the embedding is already close to a power of two, it is simply normalized and zero-padded, as when a 768-dimensional vector is loaded into ten qubits [44], [47]. When it is not, or when the qubit budget is smaller, a classical reduction such as PCA, t-SNE, or UMAP is applied first, so that the reduction and the amplitude map together fix the geometry the circuit actually sees [48]–[50].

The reason to accept this encoding is representational. Storing a large vector in a small register makes overlap- and inner-product-based operations natural, because the normalized state is the embedding. SWAP-test similarity and quantum-kernel pipelines exploit this directly, letting the geometry of the encoded state stand in for the similarity structure of the embedding space [21], [46], [51]. A distinct thread goes further and encodes complex-valued rather than real-valued amplitudes, representing tokens as complex state vectors and building complex-valued semantic operations on top of them, which uses a property of the amplitude space that has no direct classical-vector analogue [45], [52].

The signature difficulty is state preparation, and it is where the studies differ most in how honestly the cost is treated. Most rely on exact library routines that prepare an arbitrary amplitude vector, with the preparation depth absorbed into a simulator call [43], [44]. A smaller set confronts the cost directly through divide-and-conquer amplitude encoding, which trades circuit depth against ancilla width and is validated on `ibm_kyoto` for a word-sense disambiguation subset alongside the noiseless-simulation classification results [49]. Loading an arbitrary vector generally requires a

circuit whose depth can offset the qubit savings, so experiments that hide this behind high-level routines remain useful for comparing representation strategies under a fixed qubit budget but should not be read as hardware-ready evidence unless the preparation cost is included.

### C. SYNTACTIC AND DISCOCAT ENCODING

Syntactic encoding is the characteristic interface of compositional QNLP. Instead of first reducing a sentence to a flat feature vector, the model begins with a grammatical derivation. Words are assigned grammatical types, the sentence is reduced according to a pregroup or related grammar, and the resulting string diagram is translated into a tensor network or quantum circuit [17]. What makes the family a single encoding category is a shared two-stage construction: a diagram that fixes the circuit topology from grammar, followed by a rule that turns that diagram into a parameterized qubit circuit. The 29 studies in the corpus differ mainly along these two stages, and it is these differences, rather than the downstream task, that define the encoding-level variants below.

The first stage is how the diagram is obtained. The canonical route is a pregroup derivation, in which each word receives a pregroup type and grammatical reductions become the wires that connect word boxes; pregroup parsing is efficient and underlies most DisCoCat implementations [18]–[20]. In practice this parse is often produced by a broader-coverage CCG parser and then rewritten into DisCoCat form, as in studies that derive their diagrams with the Bobcat or DepCCG parsers before compilation [53]–[55]. A second variant replaces the parse with a fixed reader that imposes a uniform composition rule instead of syntax: a bag-of-words reader composes words through a single commutative spider, and a word-sequence reader composes them left to right into a tensor train, both of which appear as structure-free counterparts to the grammar-based diagrams [18], [56], [57]. These readers trade syntactic fidelity for shallower, more uniform circuits and are frequently used as structure-free reference points within DisCoCat studies. A third variant changes the scope of the diagram itself, lifting the sentence-level construction to the discourse level so that pronouns and cross-sentence reference are represented, as in DisCoCirc-style coreference models [58].

The second stage is how the diagram becomes a qubit circuit. Each atomic type is assigned a fixed number of qubits, so the noun and sentence types set the circuit width, grammatical reductions compile to Bell-style effects between wires, and function words such as relative or possessive pronouns are realized as Kronecker or GHZ spiders [20]. Word boxes are then filled by a parameterized ansatz, and the dominant choices are the IQP ansatz used by default in lambeq and, where circuit expressivity is compared, the Sim-14, Sim-15, and strongly entangling ansätze [4], [18]. Some studies design task-specific ansätze on top of the same grammatical skeleton rather than proposing a new diagram [53], [59]. A final variant intervenes between the two stages by rewriting the diagram before compilation: ZX-calculus is used to simplify the grammar-derived circuit and reduce its resource cost [60].

This category differs from vector-loading schemes. The circuit is not only a host for numerical features; its wiring reflects syntactic composition. This gives syntactic encoding a clear interpretability advantage. Grammatical roles, reductions, and word interactions remain visible in the model structure, which explains why nearly all compositional papers in the corpus use this category.

The tradeoff is flexibility. Syntactic encoding depends on parsers, grammar formalisms, and sentence structures that can be translated into manageable circuits. Many experiments therefore use short, controlled, or grammar-friendly datasets. As sentences become longer, noisier, multilingual, or document-level, the parser and circuit-growth requirements become more restrictive. Thus, syntactic encoding is the most linguistically structured category, but it is also the category that most depends on the availability of a reliable grammar-to-circuit pipeline.

### D. STRUCTURED FEATURE-MAP ENCODING

Structured feature maps encode data through circuits with phase rotations, Pauli operators, and entangling gates whose parameters depend on products or interactions of input features, so that the encoded state defines an implicit feature space in which examples are compared by state similarity [21]. Within the corpus the map is drawn from a small set of families. The most common is the ZZ feature map, which introduces pairwise feature interactions through $ZZ$ entangling rotations [61]–[63]. IQP embeddings, which apply diagonal phase gates in the Hadamard basis, are used similarly [64], [65]. A few studies construct higher-order maps from selected Pauli terms, for example combined $X$, $Y$, and $ZZ$ gates or a hierarchy of Pauli-$Y$ and second-order $YY$ interactions, in order to encode richer correlations [65], [66]. One study instead reads the feature map out through local observables, using a projected quantum kernel to obtain a lower-variance similarity estimate [67].

Because both the qubit count and the number of encoded interactions grow with the input dimension, a structured feature map is almost always preceded by an aggressive classical reduction. In practice this is usually PCA to on the order of ten to a few dozen components, and occasionally a wavelet or related projection, applied to an already compressed text representation such as a TF-IDF vector or a reduced transformer embedding [64], [68], [69]. The reduction is thus part of the encoding, since it fixes both the qubit budget and which feature interactions the map can express.

The benefit is that feature interactions are built into the map. Rather than using the circuit only to host features, the feature map defines the geometry in which examples are compared, which makes the approach attractive for testing whether a quantum kernel yields a useful similarity space for text.

The limitation is that the feature map becomes a strong modeling assumption. Performance depends on whether the

chosen interactions are relevant to the linguistic task and on the preceding compression, so a structured feature map offers a principled kernel-oriented interface whose usefulness is set jointly by map design and the reduction that feeds it.

### E. OTHER SPECIALIZED ENCODING INTERFACES

The remaining studies use mechanisms that are important but too heterogeneous to treat as major standalone families. Data re-uploading addresses the small-register setting by injecting the input features multiple times across the depth of a circuit: rather than loading the data once and then applying only trainable gates, the model alternates data-dependent rotations with trainable transformations, so expressivity grows with depth rather than with qubit count. The four corpus studies differ mainly in what is re-uploaded and where. One implements a bricklayer of dense rotation gates in which repeated feature injection, rather than one-shot angle loading, is the defining encoding choice [70]. A recurrent variant re-uploads code-token embeddings across successive quantum embedding layers, so the same feature stream re-enters the circuit at every recurrent step [71]. A third injects lexical features repeatedly into a seven-qubit NISQ classifier, explicitly trading qubit width for circuit depth [72], and a fourth re-uploads sentence-transformer embeddings across the layers of an eighteen-qubit circuit, using qubit count and depth jointly to control capacity in a quantum LLM fine-tuning setting [73]. The shared advantage is that expressivity can be raised without more qubits; the shared drawback is that the encoding burden moves from width to depth.

Hybrid embedding is a system-level interface rather than a single gate-level rule. A classical model first produces a compact representation, and a quantum layer then consumes it through an angle, amplitude, or less explicitly specified feature-injection mechanism. In the corpus the three studies are distinguished mainly by how aggressively the classical stage compresses and how clearly the quantum loading is described. One compresses fused text and image embeddings into an eight-dimensional latent vector and matches it to an eight-qubit VQC for emotion recognition in dialogue [74]. Another compresses GloVe and fastText code-token embeddings into a minimal intermediate representation before a four-qubit vulnerability classifier, without specifying the gate-level loading rule [75], and a third fuses GloVe text features with VGG16 visual features through a quantum layer for misinformation detection, again with no stated state-preparation method [76]. This underspecification is characteristic of the family and feeds its main difficulty, attribution: when most semantic processing happens in the classical stage, the independent contribution of the quantum layer is hard to isolate.

Hamiltonian and observable encoding is even more specialized. The single corpus study maps text-derived vectors onto Pauli-string Hamiltonians and reads them out as quantum observables rather than preparing them as amplitude or rotation states, which avoids conventional state encoding entirely and is reported to keep qubit and gate complexity logarithmic [77]. This relocates the loading problem from state preparation to observable design and may sidestep some state-preparation costs, but with only one study in the corpus the evidence base is too small to support a separate taxonomy family.

### F. TAXONOMIC IMPLICATIONS

The taxonomy makes clear that encoding is not a neutral preprocessing choice in QNLP. It determines which linguistic information reaches the quantum circuit, how much classical compression is required, and where the main resource bottleneck appears. Angle and amplitude encodings start from numerical text representations and differ mainly in how they trade circuit width, depth, and state-preparation cost. Syntactic encoding instead gives the grammatical analysis a structural role in the circuit. Structured feature maps define the comparison space used by kernel methods, while the specialized interfaces modify these patterns through repeated data injection, classical–quantum embedding layers, or observable-based representations.

This distinction matters because many cross-paper comparisons implicitly compare different views of the input rather than only different learning architectures. A VQC using compressed word or sentence vectors, a DisCoCat circuit generated from a parser, and a kernel model built around a designed feature map make different commitments about what should be preserved from the original text. Their results are therefore shaped jointly by the classical representation, the compression method, the encoding circuit, the ansatz, the measurement rule, and the execution backend.

Table 3 condenses this section into a single comparative view along the four axes on which the families differ most: how the qubit requirement scales with the input dimension, what the main trainability risk is, whether the corpus-typical instantiations remain classically simulable, and what part of the linguistic representation survives the mapping. Its cells restate, in compressed form, the per-family analyses of Sections IV-A–IV-E.

The taxonomy is therefore used in the remainder of the survey to organize the evidence, not to rank the encodings. The available studies do not yet show that any single family is generally superior across QNLP tasks. Instead, they show that the encoding stage frames the experiment itself: it defines the form of the linguistic input, constrains the feasible circuit resources, and conditions how performance, scalability, and quantum-advantage claims should be read.

## V. COMPARATIVE ANALYSIS OF ENCODING FAMILIES

This section uses the taxonomy to assess how far the main encoding families have developed in practice. Rather than revisiting the definition of each family, the focus is on what the corpus reveals about their use: which encodings have become standard design choices, which remain exploratory, and how strongly the reported results are supported by datasets, baselines, hardware assumptions, and scalability analysis.

**TABLE 3.** Comparative view of the encoding families along four axes: qubit scaling with input dimension $d$, dominant trainability risk, classical simulability at corpus-typical sizes, and preservation of semantic or syntactic structure. $d'$ denotes the feature count surviving classical reduction; $q$ the register width.

| Encoding family | Qubit scaling | Trainability risk | Classical simulability | Semantic preservation |
|---|---|---|---|---|
| Angle | $q \geq d'$, linear in retained features | Low: shallow rotation layers, standard parameter shift | Trivial at corpus widths (2–8 qubits) | Low: only the $d'$-dimensional remainder of an aggressive reduction survives |
| Amplitude | $q = \lceil \log_2 d \rceil$; state-preparation depth scales with $d$ | Preparation depth, routinely hidden in simulator calls | Trivial as statevector; hardware cost unpaid in most studies | High up to normalization: the full vector is the state |
| Syntactic/DisCoCat | $q$ set by sentence length and grammatical wire types, not by $d$ | Shallow per-word ansätze; parser coverage is the practical limit | IQP blocks conjectured hard; corpus instances small enough to simulate | Syntactic structure explicit in circuit topology; lexical detail limited by small ansätze |
| Structured feature map | $\approx$ one qubit per retained feature after reduction | Map is fixed, not trained; risk moves to map–task mismatch | Second-order maps conjectured hard to simulate at scale | Depends on whether hand-chosen interactions match the task |
| Data re-uploading | Fixed small $q$; depth grows with re-injections | Depth-induced gradient decay as layers accumulate | Trivial at corpus widths | Moderate: repeated injection retains more than one-shot loading at equal width |
| Hybrid embedding | Minimal $q$ (4–8); latent dimension matched to register | Attribution: classical encoder dominates learning | Trivial | Delegated: whatever the classical encoder compresses into the latent survives |
| Hamiltonian/observable | $q = \lceil \log_2 d \rceil$ with no state preparation | Single study; untested at scale | Trivial at demonstrated size | Full vector enters as observable coefficients |

The analysis is based on the corrected corpus of 125 primary studies, classified under the encoding comparison framework of Section III-F; the complete per-study classification is listed in Appendix A. Each study is assigned to its dominant encoding interface, although most implementations combine several steps before measurement: text representation, feature reduction, circuit encoding, a trainable or fixed quantum model, and a readout rule. This matters because an observed result is rarely attributable to the encoding alone. It often reflects the full pipeline in which the encoding is embedded.

Before examining the families individually, it is worth noting that the choice of encoding is not independent of the model family in which it appears. Figure 2 crosses the two dimensions, and the resulting distribution is far from uniform: most encodings concentrate in one or two model families rather than spreading evenly across them. The three dominant encodings account for 108 of the 125 studies, yet two of them are all but fixed by the architecture they serve. Syntactic/DisCoCat encoding and the compositional model family form a two-way lock, since all 29 syntactic studies are compositional and all 29 compositional studies are syntactic, and a weaker version of the same coupling ties structured feature maps to quantum kernels, where six of the nine structured-map studies live. Angle encoding, by contrast, is the general-purpose default shared across the remaining gate-model architectures, and amplitude encoding co-resides with it inside VQCs and kernel methods. Comparing "encoding families" across the corpus is therefore partly comparing the model families they are bound to. This motivates the approach of the four subsections that follow: each holds one established family fixed and asks how its use changes across the contexts in which it appears. The specialized interfaces are treated last.

### A. ANGLE ENCODING ACROSS MODEL FAMILIES

Angle encoding is the one interface that is not bound to a single architecture, so it is the natural place to begin: rather than asking which encoding each model family prefers, we hold the encoding fixed and ask how the same data-dependent rotation is repurposed as the surrounding model changes. The 51 angle-encoding studies distribute across five model families, VQCs (20), hybrid neural models (10), recurrent models (10), quantum transformers (9), and quantum kernels (2), and their treatment of the rotation differs enough that "angle encoding" names a role rather than a fixed operation. Table 4 summarizes the split; the remainder of this subsection reads it family by family. Qubit budgets are reported by 34 of the 51 studies and, with one structural exception, stay within a narrow 2–8 band, so the differences below are differences in where the rotation sits, not in how wide a register it drives.

In VQCs, angle encoding appears in its canonical form: the rotation layer is the quantum front door, followed only by a trainable ansatz and a measurement, so the encoding carries the entire burden of moving text into the circuit. This is also the family with the widest spread of classical front-ends, which is why the compression the encoding demands is most visible here. Studies load TF-IDF or bag-of-words features [78], transformer embeddings reduced by one or two orders of magnitude, from BERT's 768 dimensions to ten [36], from DistilBERT to four [37], or as far down as two values before an $R_x$ layer [79], and in a few cases handcrafted linguistic descriptors such as character entropy and diacritic counts [42]. Because the circuit is the model rather than a component of it, the VQC setting also hosts almost all of the corpus's controlled encoding comparisons: a QCNN study finds angle encoding the best of angle, amplitude, and IQP maps [80], and a sentiment study finds an $R_y$ angle map beating a Pauli feature map under matched circuits [23]. The same canonical pattern recurs across further tasks and domains in the corpus. Examples include a dressed-quantum-circuit transfer-learning classifier built on frozen BERT embeddings [81]; SBERT embeddings fused with Pauli expectation values for scholarly AI-text detection [82]; angle-encoded word-sense disambiguation on WordNet sentence-sense pairs [83]; and a privacy-oriented quantum language

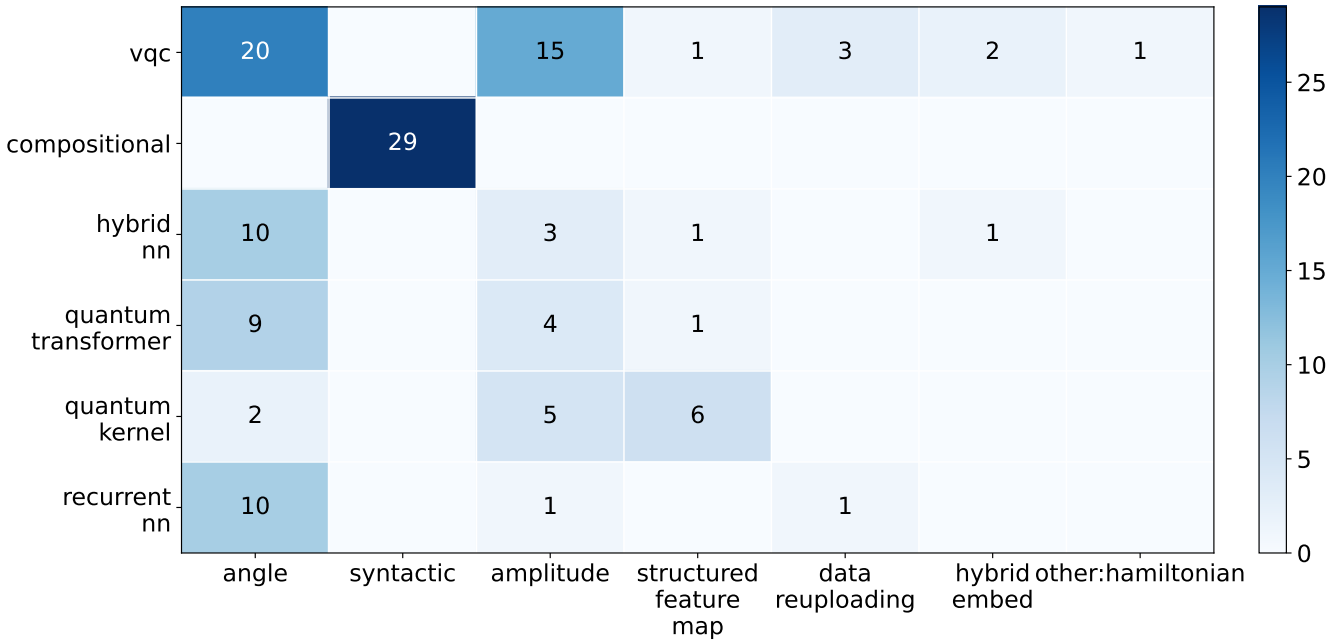



**FIGURE 2.** Encoding family by model family in the final primary corpus. Cell values report the number of studies assigned to each encoding–model-family pair.

**TABLE 4.** Angle encoding split by model family in the final primary corpus.

| Model family | Studies | Median qubits (range) | Typical classical front-end | Role of the angle rotation |
|---|---|---|---|---|
| VQC | 20 | 4.5 (2–16) | TF-IDF, BERT/SBERT/DistilBERT, handcrafted features | Sole quantum front door before a trainable ansatz |
| Hybrid neural | 10 | 4.5 (2–8) | Output of a classical transformer/RNN encoder | Submodule consuming an already-compact representation |
| Recurrent neural | 10 | 4 (4–8) | Word or BERT embeddings, per token | Data input inside the recurrent cell, re-entered each step |
| Quantum transformer | 9 | 4 (4–28) | Word embeddings, multimodal features | State preparation for quantum attention |
| Quantum kernel | 2 | 4 (4) | TF-IDF or compressed feature vectors | Definition of a similarity-circuit geometry |

model trained on NISQ hardware and transferred to classical downstream tasks [84]. Further instances are post-hoc calibration of language-model confidence via a variational circuit [85]; a quantum depthwise-convolutional classifier that pairs amplitude-encoded word embeddings with an angle-encoded sentence representation [86]; noise-robust multimodal sentiment analysis trained with an SPSA optimizer [87]; and an all-quantum LSTM with a self-attention block whose noise robustness is tested explicitly [88]. The attribution of any result is comparatively clean here, but at the price of a circuit that only ever sees a heavily compressed view of the input.

Hybrid neural models move the rotation downstream. The angle layer is no longer the front door but a submodule embedded in a larger classical network, so a transformer or recurrent encoder performs most of the representational work and the quantum circuit consumes a representation that has already been compressed classically. This is visible both in the front-ends, MARBERT, AraBERT, or XLM-R embeddings compressed into shallow circuits whose Pauli readouts are fused back into the classical model [38], a BERT-and-GRU multimodal pipeline projected into a two-qubit circuit, a transformer generator whose tokens are injected as rotations [89], and in the registers, which are the smallest in the angle subcorpus. The same downstream-submodule role recurs in further hybrid designs. These include a quantum fuzzy neural network layered on top of angle-encoded features for noise-robust sentiment analysis [90]; quantum tensor-network fusion of text, audio, and visual streams for multimodal sentiment [91], [92]; a compact variational embedding layer used as a drop-in replacement for a classical word-embedding layer [93]; a quantum adaptation of the Mamba selective state-space architecture for language modeling [94]; and a self-supervised, GLUE-benchmarked quantum-classical BERT variant pre-trained under an explicit noise model [95]. The cost is the mirror image of the VQC case: because the classical encoder dominates, the independent contribution of the quantum rotation is the hardest to isolate of any family.

Recurrent models repurpose the rotation temporally rather than spatially. Here angle encoding is applied inside a quantum recurrent cell, a QLSTM or QRNN block, so the same rotation interface is re-entered at every step of the sequence

rather than once at the input [24], [25], [29], [39]. This keeps the register deliberately narrow, a median of four qubits and never more than eight, because width is fixed by the cell while information enters through repeated application along the sequence, which brings this usage closer in spirit to data re-uploading than to a single loading step. The $R_y$-based AngleEmbedding is the recurring choice [24], and the family is the one most oriented toward token-level tasks such as part-of-speech tagging [96] alongside sequential sentiment classification, code-mixed part-of-speech tagging across three Indian language pairs [97], recency-weighted mental-health-aware sentiment monitoring [98], frozen-BERT emotion recognition [99], and movie-review sentiment classification competitive with classical RNN, GRU, and LSTM baselines [100].

Quantum transformers use the rotation to prepare the states that attention then compares, and this is the one place where the usual qubit discipline breaks. Features are mapped into per-token quantum states whose overlaps play the role of attention scores, so register width scales with sequence and embedding size rather than with a compressed feature count. QNet assigns one qubit per token–embedding-dimension pair and splits axes by role, with $R_x$ carrying the embedding value and $R_z$ a positional phase [26]. A near-term self-attention study reaches 28 qubits, the widest register in the entire angle subcorpus [101]. Other studies keep the per-token construction but vary what the attention operates over: a quantum self-attention network built on the QSANN architecture [102], a quantum graph transformer that folds self-attention into a graph message-passing framework [103], a mixed-density-matrix formulation of quantum self-attention [104], and a noisy-simulation image-text quantum transformer that fuses a text PQC with a vision-transformer PQC [105]. Most implementations avoid this blow-up by keeping the quantum attention block small [34], [106], but the family is nonetheless the only one whose structural logic pushes angle encoding beyond the near-universal $4$–$8$ qubit band.

Quantum kernels are represented by only two angle studies, and the contrast with the previous families is instructive. When angle encoding does appear in a kernel method, the rotation defines a similarity-circuit geometry rather than feeding a trainable ansatz [31], [107]; but kernel methods overwhelmingly prefer purpose-built structured feature maps (Section V-D), so angle encoding is a marginal, borrowed choice here rather than a native one. This is the exception that confirms the pattern: angle encoding travels everywhere, but in the one family whose encoding is itself the model, a different interface wins.

Taken together, the split shows that holding the encoding fixed does not hold the pipeline fixed. The same rotation is a complete front door in VQCs, a downstream submodule in hybrid models, a per-step cell input in recurrent models, an attention-state preparation in quantum transformers, and a borrowed similarity map in kernels. These roles differ in exactly the dimensions that matter for evaluation: how much of the model the encoding accounts for, whether register width is set by compression or by architecture, and how cleanly the quantum contribution can be attributed. The one invariant across all five families is the heavy classical compression that precedes the circuit. This confirms the encoding–model-family correlation of Fig. 2 from the opposite direction: even when the encoding is the same, it is the model family that determines what the encoding does, which is why angle-encoded results are not directly comparable across architectures despite sharing a label.

### B. AMPLITUDE ENCODING ACROSS MODEL FAMILIES

Amplitude encoding is the natural counterpoint to angle encoding, and reading its 28 studies through the same per-family lens makes the contrast concrete. It appears in the same host families, VQCs (15), quantum kernels (5), quantum transformers (4), hybrid neural models (3), and a single recurrent model, but with an almost inverted profile, both in which families it favors and in the resources it consumes. Where angle encoding compresses text to a handful of rotation parameters over a $2$–$8$ qubit register, amplitude encoding loads a full normalized vector into the $2^n$ amplitudes of an $n$-qubit state, so its registers cluster near $\log_2$ of the embedding dimension rather than at a fixed small width. Table 5 summarizes the split, and the immediate signal is the qubit column: every family sits at a median near ten qubits, against the median of four seen for angle encoding, because the encoding is chosen precisely to avoid the aggressive projection that angle encoding requires.

VQCs are again the dominant host, but the role differs from the angle case. Amplitude encoding here prepares the input state that a trainable ansatz—typically EfficientSU2 or RealAmplitudes—then classifies [48]. The front-end is almost exclusively a dense contextual embedding normalized and padded to the nearest power of two: a 768-dimensional SBERT vector into ten qubits [44], BERT or ELECTRA embeddings padded from 768 to 1024 amplitudes over ten qubits [43], and mBERT embeddings reused across multilingual transfer [47]. The $\log_2$ packing is stated as the point of the method rather than an implementation detail [108]. Where a study compares the two interfaces directly, it frames amplitude encoding as the high-fidelity option and angle encoding as the cheaper fallback: one sentence-classification model reports its amplitude variant at $0.81$ accuracy against $0.71$ for an MLP-reduced angle variant of the same system [44]. The same host role recurs across further tasks. Examples include an attention-augmented amplitude classifier evaluated on a non-standard Kaggle text corpus [109]; an Electra-based amplitude transfer-learning system for Italian-language sentiment analysis [110]; and an amplitude-encoded QRNN with ancilla-mediated state transfer applied to code-mixed social-media text [111]. Further instances are a complex-amplitude variational topic model that reduces parameter count relative to a classical neural topic model while improving topic coherence [112]; a BERT-to-nine-qubit amplitude pipeline for smishing and spam-text detection [113]; and a ten-qubit amplitude classifier applied to word-sense disambiguation in

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**TABLE 5.** Amplitude encoding split by model family in the final primary corpus.

| Model family | Studies | Median (range) qubits | Typical classical front-end | Role of the amplitude state |
|---|---|---|---|---|
| VQC | 15 | 10 (9–16) | Normalized BERT/ELECTRA/DistilBERT/SBERT embeddings | Input state prepared for a trainable ansatz |
| Quantum kernel | 5 | 10.5 (7–12) | Word/sentence embeddings, TF-IDF vectors | Overlap geometry read out by a SWAP test |
| Quantum transformer | 4 | 7.5 (4–12) | Word or contextual token embeddings | State preparation for quantum attention |
| Hybrid neural | 3 | 9 (6–12) | Multimodal or token embeddings | High-fidelity submodule after a classical encoder |
| Recurrent neural | 1 | — | Chinese word embeddings | Per-gate state inside a decomposed QLSTM |

a low-resource ancient language [114]. This family is also markedly more recent than its angle counterpart, with every VQC amplitude study appearing from 2024 onward, which tracks the point at which transformer embeddings became the default text representation and made a dense-vector encoding worth its qubit cost.

Quantum kernels are where amplitude encoding is most at home, in contrast to the marginal, borrowed status angle encoding held there. When a normalized embedding is amplitude-encoded, the inner product of two states equals the inner product of the underlying vectors, so a SWAP test or state-overlap estimate reads out cosine-like semantic similarity directly, with no training involved [45], [51], [115]. This is why the semantic-similarity task cluster of the corpus lives almost entirely in this cell, and why kernel studies apply the same construction to fastText, GloVe, or BERT embeddings and to TF-IDF vectors feeding a quantum nearest-neighbor classifier [46], [116]. The fit is principled rather than incidental: the encoded state is the geometry the kernel measures, and some studies push the density further with complex-amplitude variants that map pairs of dimensions to the real and imaginary parts of each amplitude, halving the register again [45].

Quantum transformers use amplitude encoding, as they use angle encoding, to prepare the states that attention compares, but over noticeably smaller and more variable registers than in the VQC case [117]–[121]. Amplitude encoding is rarer here than angle encoding (four studies against nine) because preparing an arbitrary state for every token is more expensive than injecting a rotation. The studies that do adopt it tend to encode a single pooled or sentence-level vector rather than a per-token sequence. Hybrid neural models place amplitude encoding in the same downstream-submodule role that angle encoding occupies there, but selected explicitly for fidelity. One knowledge-graph study contrasts an amplitude model chosen for high-dimensional fidelity against an angle model chosen for gate efficiency—the trade-off between the two families stated within a single system [122]. A multimodal model amplitude-encodes fused visual and textual features [123], and a space-efficient hybrid architecture amplitude-encodes representations for natural-language-to-Prolog query translation, discussing the state-preparation cost explicitly as a limitation [124].

The recurrent family is the sharpest point of the contrast. Amplitude encoding appears in exactly one recurrent study, a decomposed quantum LSTM over Chinese word embeddings [125], against the ten recurrent studies that use angle encoding, and the asymmetry is not accidental: re-preparing an arbitrary amplitude state at every step of a sequence is prohibitively expensive, whereas re-applying a rotation inside a cell is cheap. The encoding that is native to recurrent processing is therefore the opposite of the one that is native to overlap-based comparison. Read against the angle-encoding split, amplitude encoding partitions the same families by an opposite logic: angle encoding minimizes width and pays for it in classical projection, while amplitude encoding minimizes projection and pays for it in register width and, more importantly, in state-preparation depth. That last cost is the one most often abstracted away by simulator calls, so amplitude-encoded results are the ones whose reported qubit efficiency is most sensitive to whether the price of preparing the state has actually been counted.

### C. SYNTACTIC/DISCOCAT ENCODING ACROSS ANSÄTZE

Syntactic/DisCoCat encoding cannot be split by model family the way the previous two families were, because all 29 of its studies belong to a single model family: compositional QNLP. This follows from the definition given earlier, since the grammatical reduction that wires the circuit is the model, so there is no separate architecture to vary. What does vary, and what plays the structural role model family played above, is the ansatz used to turn each grammatical box into gates, because that choice is the one genuine degree of freedom left once the parser has fixed the diagram. Two properties are shared across the whole subcorpus regardless of ansatz. First, the toolchain is a near-monoculture: 22 of the 29 studies build their circuits with lambeq, and most of the rest use the related DisCoPy/pytket stack, in contrast to the diverse PennyLane, Qiskit, and TorchQuantum backends seen for the vector-loading families. Second, qubit count is not a function of the ansatz: it is fixed by the diagram, growing with sentence length and the number of grammatical wire types regardless of which gate template is dropped onto each box. Table 6 groups the studies by ansatz, and the qubit column shows exactly this: the reported range is nearly identical in the IQP bucket and in the small named-alternative bucket, two to twenty-five against four to ten, because register width tracks the sentence, not the gate design.

The IQP ansatz is the family's workhorse and accounts for two-thirds of the subcorpus. It originates in the two "generic families" of ansätze proposed alongside the first NISQ demonstrations: a CNOT-and-$U(3)$ construction and an IQP construction of alternating Hadamard and controlled-

**TABLE 6.** Syntactic/DisCoCat encoding split by circuit ansatz in the final primary corpus.

| Ansatz family | Studies | Median qubits (range) | Gate-level construction | Noun parameterization |
|---|---|---|---|---|
| IQP (LAMBEQ default) | 19 | 7 (2–25) | Hadamard layers alternated with layers of parameterized controlled-$R_z$/CZ rotations, one block per grammatical reduction | Nouns are trainable rotations, same as other word types |
| Named alternative (SimpleSA, GeneralQC, DCA) | 3 | 5 (4–10) | $H$–CNOT–$R_z$–$H$ or $H$–$CR_x$/$R_x$–CNOT/$R_x$–$CR_z$ blocks | Nouns fixed as identity gates or frozen embedding-derived states |
| Multi-ansatz benchmark (Sim14/15, StronglyEntangling, MPS/Spider/Tensor vs. IQP) | 5 | 13 (7–20) | Same diagram compiled through several interchangeable circuit or tensor-network templates, compared head-to-head | Nouns trainable in every compared variant |
| ZX-calculus/underspecified | 2 | 10 (single report) | Diagram or grammar tree compiled to a circuit via a generic, unlabeled routine; gate template not stated | Not reported |

phase layers [19]. The IQP variant was then formalized as a depth-$d$ word-circuit template—arity-one words as a single Euler rotation, higher-arity words as $d$ layers of Hadamard-plus-neighboring-controlled-$R_z$ gates—motivated explicitly by IQP's conjectured classical hardness [20]. It was tuned further with a cheaper single-rotation option for low-arity words [18]. This is the construction that lambeq packages as its default `IQPAnsatz` class [126]. The great majority of application papers in the corpus adopt it unmodified, changing only what is fed into the diagram or read out of the circuit. Examples include video-caption classification [55]; multimodal captioning with a separate image encoder [127]; knowledge-graph relation modeling [128]; sentiment classification that keeps the same `IQPAnsatz` for its quantum runs while using a Spider tensor-network ansatz only as a classical simulation baseline [129]; Hindi sentiment classification [130]; and natural-language inference with tied parameters [131]. Further instances are optimizer benchmarking at fixed ansatz depth [132], [133]; an ensemble over five diagram-construction readers that all compile to the same IQP circuit [134]; a hand-built IQP-equivalent circuit for pronoun resolution built from Fock-space projections [58], [135]; a direct translation of the same block structure to Chinese grammar [136]; an English–Persian translation pipeline that Euler-decomposes noun rotations [137]; and a musical-phrase model that applies the same Euler decomposition to snippet "words" [138]. The one recurring modification within this bucket is additive rather than structural, such as a dedicated negation gadget layered on top of an otherwise standard IQP circuit [54], which confirms that once fixed by the toolkit, the ansatz itself is treated as infrastructure rather than as a variable most studies think to manipulate.

A second, much smaller cluster treats the ansatz as the thing to redesign, and does so along one consistent axis: removing the noun from the pool of trainable gates. SimpleSA replaces the IQP block with an $H$–CNOT–$R_z$–$H$ construction in which noun boxes are fixed identity gates rather than trainable unitaries, and reports higher accuracy than IQP with fewer parameters [59]. GeneralQC generalizes the idea to three concrete gate layouts: $H$-$CR_x$, $R_x$-CNOT, and $R_x$-$CR_z$. The two best-performing variants, $R_x$-CNOT and $R_x$-$CR_z$, are again the ones that leave nouns non-parameterized, reaching 96.91% accuracy against a 79.38% IQP baseline [53]. Parameter count does not track this pattern, however: both winning layouts use more trainable parameters than the IQP baseline (109 and 218 against 91), not fewer. The discrete circuit ansatz applies the same principle by a different mechanism, replacing the noun's trainable box with a fixed state derived from a classical word embedding through amplitude-style state preparation, and adding only a small trainable rotation keyed to the grammatical type, specifically to handle words the parser has never seen [139]. Across all three, the shared premise is that a word's syntactic role, not its lexical identity, is what needs to be learned. Freezing the noun consistently improves or preserves accuracy on the tasks reported. It also cuts parameter count in SimpleSA specifically, but GeneralQC's best layouts show this need not hold in general: a redesigned ansatz can spend more parameters elsewhere in the circuit while still gaining from non-parameterized nouns.

A third cluster asks the ansatz question directly, by fixing the diagram and swapping the gate template. The widest comparison pits lambeq's circuit ansätze, IQP, StronglyEntangling, Sim14, and Sim15, against its tensor-network ansätze, MPS, Spider, and plain Tensor, and finds Sim14 and Spider narrowly ahead of IQP among the circuit and tensor families respectively, with Sim15 essentially matching IQP and only StronglyEntangling clearly behind [4]. A second comparison of IQP against the Sim-et-al. "circuit 14/15" and StronglyEntangling ansätze finds the opposite depth dependence: StronglyEntangling is ahead at shallow depth, but IQP and Circuit14 converge faster and overtake it as circuit depth increases [140]. A related four-way comparison across six datasets gives IQP the best mean accuracy overall, while StrongEntangling wins on the hardest of them [57]. The same authors elsewhere present Sim14 and the Spider ansatz only as illustrative options, without running the comparison [141]. A narrower, non-lambeq comparison pits IQP against Sim14 directly on SQL circuits and finds Sim14 slightly more accurate but harder to train, concluding that IQP's simplicity is the better default for that task [142]. Read together, these head-to-head studies complicate rather than confirm the workhorse status IQP holds by default elsewhere in the corpus. IQP is competitive rather than dominant: a hardware-efficient or tensor-network alternative beats it on accuracy or convergence speed in most, though not all, of the papers that run the comparison. It is also the ansatz that pulls ahead as circuit depth grows in the one study that varies

depth explicitly. Its dominance in the rest of the subcorpus therefore reflects toolkit convenience at least as much as a demonstrated performance advantage.

Two studies leave the gate template unspecified. One routes the diagram through ZX-calculus rewriting before compiling it to a circuit, but delegates the compilation to a generic diagram-to-circuit call without naming the resulting gates [60]; the only concrete claim recoverable from the paper is that ZX-simplification reduces the qubit count needed for a given sentence. The other builds a custom, non-lambeq pipeline and shows circuit diagrams with parameterized boxes at each atomic type, without labeling the gate template they instantiate [56]. Neither paper permits the ansatz to be classified beyond "some parameterized construction," which is itself informative: it marks the two studies in the subcorpus with the least methodological transparency about the one design choice this section is organized around.

Read together, the split shows that syntactic/DisCoCat encoding is more homogeneous at the ansatz level than at the application level: a single IQP-derived template, inherited from the family's foundational papers and packaged as lambeq's default, is adopted essentially unmodified by two-thirds of the subcorpus regardless of whether the downstream task is sentiment, translation, or music. The two genuine departures pull in different directions: one strips the ansatz down by freezing nouns to cut parameters, the other benchmarks it against hardware-efficient and tensor-network alternatives and repeatedly finds it beatable, and together they show that the field's default gate template is a convenience inherited from tooling rather than a demonstrated optimum. What the ansatz choice does not touch, in any of the four buckets, is qubit count, which stays governed by the grammar diagram itself [60], [142].

### D. STRUCTURED FEATURE-MAP ENCODING ACROSS CONSTRUCTIONS

Structured feature-map encoding is the smallest of the four families addressed here, with nine studies, and it repeats the concentration seen for syntactic encoding: six of the nine are quantum-kernel methods, with only single instances in VQC, hybrid neural, and quantum-transformer models. Because model family again barely varies, the dimension that plays its role is the feature-map construction itself. That axis is natural for this family in any case. Unlike angle and amplitude encoding, which load feature values, a structured map builds feature interactions into the circuit through phase rotations and entangling gates; the choice of map is therefore a modeling assumption about which interactions matter. Two properties frame the split. Like angle encoding, the family depends on aggressive classical compression before the circuit: the high-dimensional text representation is reduced by PCA or a projection layer first. Unlike angle encoding, register width then tracks the number of retained features at roughly one qubit each. This is why the reported registers are comparatively wide, with a median of $10.5$ qubits and a range of $5$ to $20$. Table 7 groups the studies by the map they use.

ZZ and IQP maps are the default construction, accounting for four of the nine studies. These are second-order maps that entangle pairs of features through data-dependent phase gates, and they are most at home in quantum kernels, where the encoded state defines the similarity geometry that the kernel measures: a ZZFeatureMap QSVM over a twenty-qubit register [61], an IQP feature map applied to reduced AraBERT embeddings [64], and IQP/ZZ maps feeding projected quantum kernels [67]. The same construction also appears once outside a kernel, as the primary quantum encoding layer of a hybrid model whose BERT/T5 embeddings are projected into a five-qubit ZZ map [62], which illustrates the family's borderline status: even its standard map is used both as a kernel definition and as a swappable layer inside a trainable network.

General Pauli maps, in three studies, are the more expressive and more experimental construction. These use higher-order or non-$Z$ Pauli strings and treat the exact gate content as a design choice to be tuned. A question-answering VQC selects a specific $X/Y/ZZ$ combination after a small search [66]. A quantum-enhanced SVM adopts a Pauli-$Y/YYY$ feature map with a rotation factor over fourteen qubits produced by PCA [65], alongside an earlier Pauli-map sentiment kernel [68]. This is exactly where the family's central risk becomes visible: because the map fixes which feature interactions are represented, performance depends on whether that hand-chosen structure happens to match the linguistic task, and the map is a strong assumption rather than a learned one. The remaining two studies report only an unnamed "structured feature map", one as a QSVM branch compared against a VQC [63] and one as a feature stage inside an aviation-text transformer [69], a reporting gap that makes them hard to place against the named constructions.

Taken together, structured feature maps are the most-benchmarked yet least-dominant of the established families. Because the map is usually presented as one candidate encoding among several, the studies that use it tend to compare it directly with the vector-loading families. Those comparisons do not favor it: the group behind the Pauli-$Y$ kernel above later reports, in the controlled study cited in the angle-encoding analysis, that a simple $R_y$ angle map outperforms a Pauli feature map under matched circuits [23]. The concentration in kernels is therefore the coherent core of the family, the one setting where the map is the model rather than a replaceable layer, while the scattered VQC, hybrid, and transformer instances are experiments in borrowing a designed geometry into a trainable pipeline. This leaves the family with a narrow but principled niche, testing whether an engineered similarity space helps for text, subject to a double dependency the other families do not share: its usefulness rests on both the feature-map design and the compression stage that precedes it.

### E. EMERGING AND SPECIALIZED INTERFACES

The residual group of eight studies is best read not as one family but as three distinct mechanisms, and unlike the four

**TABLE 7.** Structured feature-map encoding split by map construction in the final primary corpus.

| Map construction | Studies | Median qubits (range) | Interactions encoded | Host families and role |
|---|---|---|---|---|
| ZZ / IQP maps | 4 | 10 (5–20) | Pairwise, second-order feature products via entangling phase gates | Mostly kernel geometry; one hybrid encoding layer |
| General Pauli maps | 3 | 11 (6–14) | Higher-order and non-$Z$ Pauli-string interactions | Kernel and VQC; gate combination chosen by search |
| Unnamed structured maps | 2 | not reported | Unspecified feature interactions | QSVM branch; transformer feature stage |

established families it cannot be split by model family for a different reason: it is already almost mono-cultural in architecture. Seven of the eight studies are variational or recurrent circuits, with a single hybrid-neural case and no kernel, compositional, or transformer instances at all. What varies, and what organizes this subsection, is the specialized mechanism itself, because each one is best understood as a targeted patch for a specific limitation of the mainstream encodings analyzed above rather than as a general-purpose alternative to them. Table 8 groups the studies by mechanism and names the trade each one makes.

Data re-uploading is the response to the width problem that constrains angle and amplitude encoding. Instead of loading features once and then applying only trainable gates, it re-injects the same feature stream across the depth of the circuit, alternating data-dependent rotations with trainable transformations, so a small register can carry more information than a single loading step would allow. The four studies make this trade explicitly: a bricklayer data-reuploading classifier over six qubits [70], a lexical-feature circuit that trades qubit width for repeated injection [72], a recurrent variant that re-uploads code-token embeddings across recurrent quantum layers for vulnerability detection [71], and a sentence-transformer pipeline that spreads a large embedding across depth over eighteen qubits [73]. The register range, from six to eighteen qubits, reflects exactly this: width is no longer minimized to a fixed small band as in angle encoding, but neither does it grow to $\log_2$ scale as in amplitude encoding, because the burden has been shifted into circuit depth instead.

Hybrid embedding is the response to the high-dimensionality problem, and it is a system-level interface rather than a gate-level rule. A classical encoder, an autoencoder or a multimodal network, first compresses the text, and sometimes other modalities, into a compact latent vector, which a small quantum layer then consumes. This is why these are the smallest registers in the specialized group, a median of six qubits: an eight-dimensional latent from a text-and-image emotion model mapped to eight qubits [74], a minimal intermediate representation of code-token embeddings over four qubits [75], and a multimodal misinformation pipeline that enriches GloVe and visual features through a quantum fusion layer [76]. The recurring theme, and the recurring weakness, is that all three are multimodal or code-analysis systems in which the classical encoder does most of the representational work, so the independent contribution of the small quantum tail is the hardest of any interface to isolate, which is the same attribution problem seen for hybrid-neural angle encoding, now taken to its limit.

Hamiltonian and observable encoding is the most conceptually distinct of the three, and it is the response to the state-preparation problem that undermines amplitude encoding's qubit savings. A single study maps text-derived data to Pauli-string Hamiltonians used as observables rather than preparing a quantum state at all, which sidesteps the state-preparation circuit entirely while keeping a logarithmic qubit count, demonstrated over nine qubits for sentiment classification [77]. This changes the object that carries the linguistic information, from a prepared state to a measured observable, and so directly targets the cost that Section V-B identified as amplitude encoding's hidden liability. With only one study, however, it is an idea rather than an established interface.

Taken together, the specialized interfaces confirm the reading of the established families from the outside in: each is defined by the mainstream limitation it tries to escape, data re-uploading trading width for depth, hybrid embedding trading quantum responsibility for classical preprocessing, and Hamiltonian encoding trading state preparation for observables. Their shared confinement to variational and recurrent circuits shows that this experimentation is happening entirely within the trainable-circuit world, not in the kernel or compositional families whose encodings are structurally fixed. The evidence, however, is thinner and less comparable than for any established family: these studies differ not only in encoding but in where the quantum component sits and how much work precedes it. The natural questions—whether depth is worth reduced width, whether the quantum layer adds value beyond the classical encoder, and whether observable encoding beats state preparation—remain open on the current corpus.

## VI. TEMPORAL EVOLUTION OF ENCODING AND MODEL CHOICES

Every family and mechanism examined so far has been analyzed as a cross-section of the corpus, encoding against model family, encoding against ansatz, at a single frozen point. This section reintroduces the one dimension held constant throughout, publication year, and asks not only how large the field has grown but why its composition changed as it grew, drawing on the mechanisms already identified in Sections IV and V to interpret the trend rather than merely report it. The final corpus spans six publication years, from two studies in 2021 to a peak of 41 in 2025. The last calendar year is included at 27 studies but was not complete at the time of writing, since the search preceded the year's close. Any

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**TABLE 8.** Specialized encoding interfaces split by mechanism in the final primary corpus.

| Mechanism | Studies | Median qubits (range) | Core mechanism | Host families and trade |
|---|---|---|---|---|
| Data re-uploading | 4 | 7 (6–18) | Same features re-injected across circuit depth, alternating with trainable gates | VQC, recurrent; trades qubit width for depth |
| Hybrid embedding | 3 | 6 (4–8) | A classical encoder emits a compact latent vector consumed by a quantum layer | VQC, hybrid; trades quantum role for classical preprocessing |
| Hamiltonian/observable | 1 | 9 | Data mapped to Pauli-string Hamiltonians or observables, with no state preparation | VQC; trades state encoding for observable encoding |

apparent 2026 decline is therefore a search-cutoff artifact rather than a documented contraction, and it is treated as such throughout.

### A. GROWTH AND THE CHANGING ENCODING MIX

Figure 3 decomposes the field's fivefold growth by encoding family, and the shape of that growth points to a specific mechanism rather than to uniform diversification. Syntactic/DisCoCat encoding, the only family represented in 2021, accounted for its entire output that year and still half of it in 2022, but its share falls monotonically after that, to 38% in 2023, 19% in 2024, and 11% by 2026, even while its absolute count stays roughly flat at three to eight studies a year. Angle and amplitude encoding absorb almost all of the volume that arrives after 2023. Angle rises from a quarter to roughly half of the yearly corpus. Amplitude, after an early spike to a quarter of the small 2022 corpus and a near-total drop to under 5% in 2023, settles into a stable quarter to third of every year from 2024 onward.

The most plausible driver of this split is a difference in barrier to entry rather than in explanatory adequacy. This is the review's own interpretation of the shift, offered because the corpus-wide counts support it structurally; no paper in the corpus frames its encoding choice explicitly in these terms, so the mechanism should be read as an inference from the aggregate pattern, not a documented finding. A DisCoCat study requires committing to a grammar formalism, a parser capable of producing pregroup or CCG derivations, and the narrow lambeq/DisCoPy toolchain that Section V-C showed also constrains the ansatz. The payoff is a linguistically interpretable wiring, but the cost is paid again on every new sentence, language, or domain. This is why the same section found the family's practicality to fall away the further the input drifts from a short, curated English sentence. Angle and amplitude encoding impose no comparable commitment: either accepts any fixed-length numeric vector. Once general-purpose pretrained text embeddings became a routine input—available to a researcher with no linguistic-formalism background at all—the marginal cost of a new study dropped to whatever off-the-shelf VQC or kernel recipe was already circulating in the broader quantum-machine-learning literature. Section V-B located that shift concretely: every VQC-hosted amplitude study from 2024 onward runs on dense transformer embeddings, and 2024 is the year amplitude's yearly share stabilizes in Figure 3. The remaining families fit the same account from the other side. Structured feature-map encoding stays in the single digits every year, apart from a one-off spike to 14% in 2023, and otherwise never moves far from three to eight percent. The three specialized interfaces—data re-uploading, hybrid embedding, and Hamiltonian/observable encoding—together never exceed one-tenth of any year's output. Each still demands bespoke, non-templated engineering (a decomposed recurrent layer, a co-designed classical encoder, a custom observable construction) that scales with the number of researchers willing to do that engineering rather than with the size of the field, so their share cannot expand the way a copy-paste encoding's can. The field's growth, in short, has followed the path of least engineering resistance: the encodings that ask the least of a newcomer, not the ones judged most linguistically or computationally motivated in Section IV.

### B. FROM COMPOSITIONAL MONOCULTURE TO ARCHITECTURAL PLURALISM

Figure 4 tells the same story from the architecture side, and because compositional QNLP and syntactic encoding are the same 29 studies (Section V-C), its decline is not a second, independent observation but the identical erosion seen from a different axis: compositional models are the entire 2021 corpus, half of 2022, and down to 11% by 2026. What replaces that share, however, is not one architecture but four, and the shape of the replacement is again best explained by portability rather than by a documented technical advantage. VQCs, the largest beneficiary, grow from an eighth of the 2022 corpus to about a third by 2025 and just under half by 2026. A VQC is not a QNLP-specific construction; Section V established it as the default host for both angle and amplitude encoding precisely because it asks nothing of the input beyond a numeric vector. Its rise is the model-family expression of the same embedding-commoditization pattern identified above. Quantum transformers and quantum kernels, by contrast, hold a steady, unspectacular seven-to-fifteen percent share from 2022 onward, with quantum kernels dipping just under that band to 7.4% in the partial 2026 count. Both require a deliberately quantum-native design decision—a state-overlap readout for kernels, an attention-like circuit structure for transformers—that cannot simply be copied from a generic VQC recipe. Their growth therefore tracks the smaller, slower-growing population of studies willing to make that architectural commitment rather than the field's overall volume.

Hybrid neural models are the one genuine outlier: negligible through 2024, then 27% of the 2025 corpus—the single

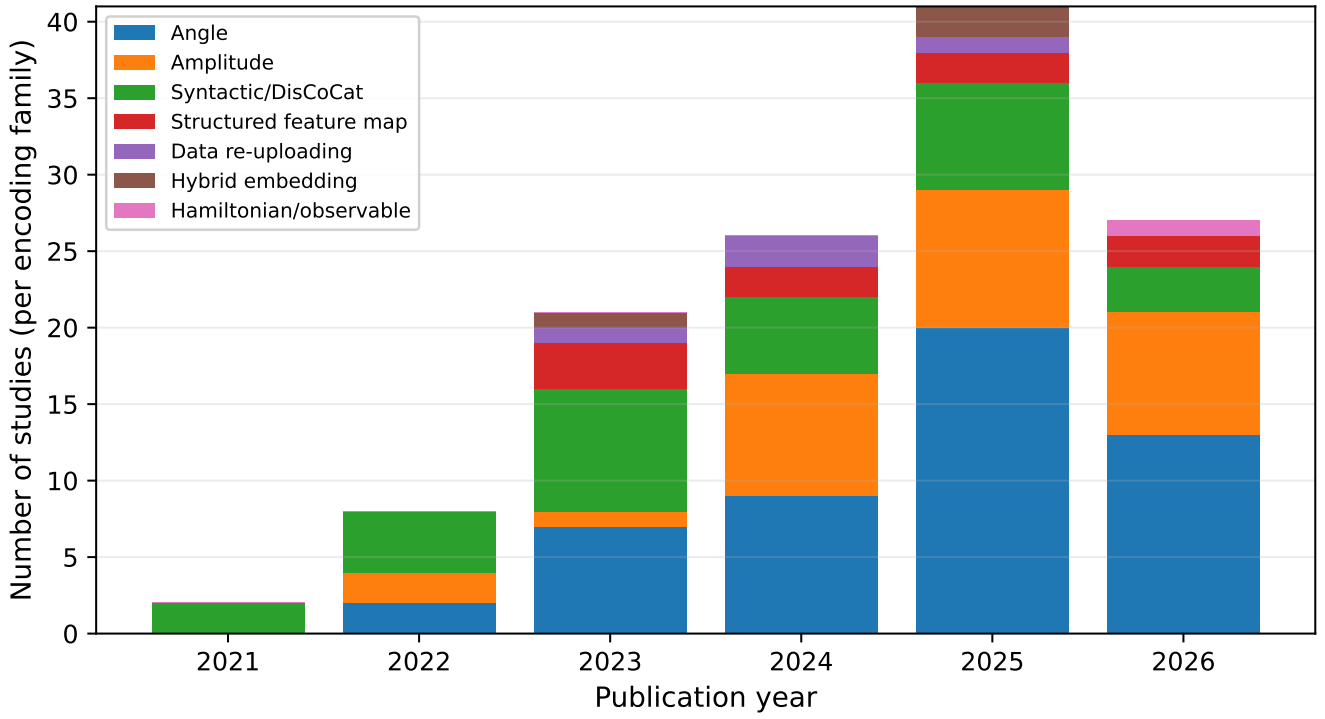



**FIGURE 3.** Encoding family by publication year in the final primary corpus (2026 is a partial year at the time of the search).

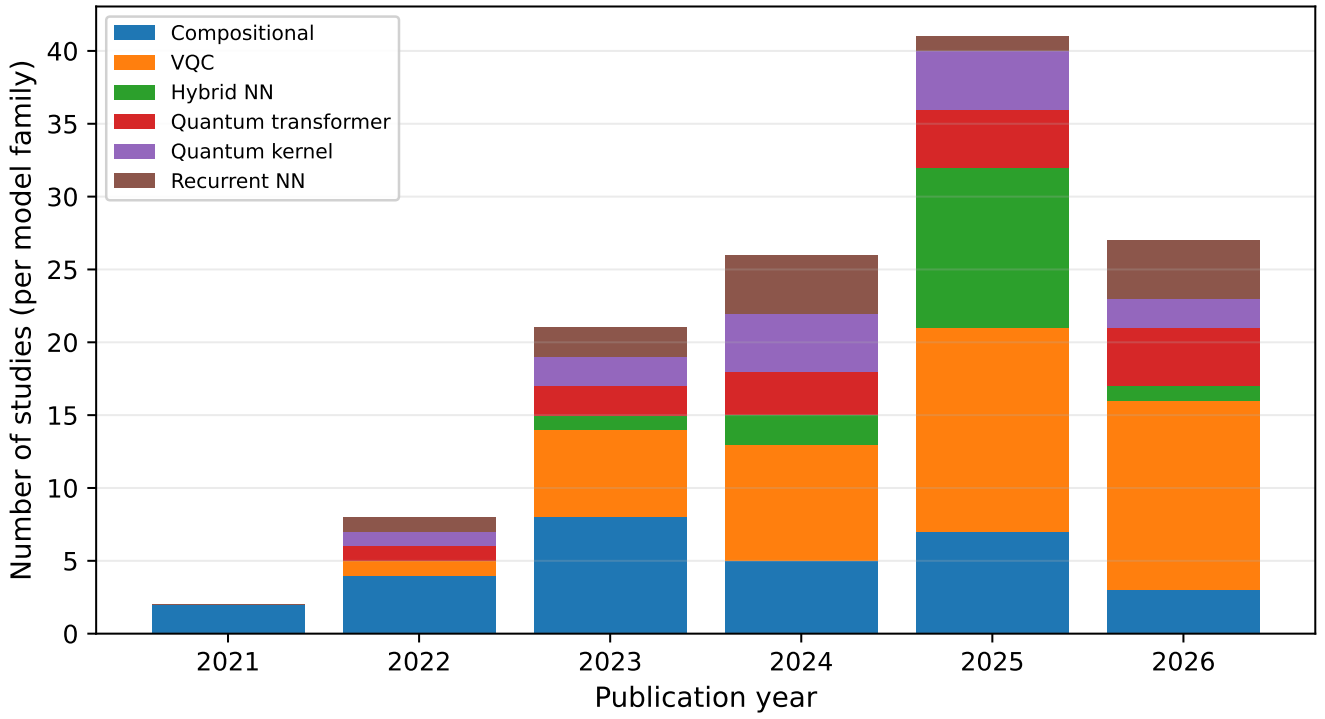



**FIGURE 4.** Model family by publication year in the final primary corpus (2026 is a partial year at the time of the search).

largest jump of any family in any year—before falling back to 4% in the partial 2026 count. Because 2026 is incomplete,

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the fall cannot yet be read as a reversal. The 2025 spike itself is real, and it is consistent with the same low-barrier mechanism taken one step further. A hybrid neural model, by construction, puts most of the representational burden on a pretrained classical encoder and reserves only a small quantum submodule; it is the architecture that asks the least genuinely quantum design work of any family in the corpus. It is also the family Section V-E flagged as hardest to attribute to the quantum component, since the classical encoder does most of the work. Read together, the two observations support the explanation already offered for the encoding-level trend: the field's fastest-growing segments are consistently those with the lowest entry cost for a researcher bringing a pretrained classical model and a small amount of quantum machinery, not those built around a bespoke quantum-native design.

## VII. TASK-LEVEL FIT BETWEEN ENCODING, MODEL FAMILY, AND REPORTED ADVANTAGE

The preceding sections asked which encoding a study uses and when; this one asks whether the choice ever pays off in a way that tracks the task itself. Every primary study in the corpus records whether it claims a quantum advantage over its classical baseline, so the question can be asked directly. Within a given task family, does any encoding or model family claim that advantage more often than the others? And if so, is there a reason to believe it, or is it an artifact of who happens to publish in that niche?

Figure 5 sets the stage: sentiment and other text classification together absorb the bulk of the corpus and every encoding family with any following at all, so before asking about advantage it is worth asking whether these two task families offer a fair test in the first place. They do not, in any strong sense. Within sentiment and other classification, claimed advantage is scattered fairly evenly across angle, amplitude, and syntactic encoding, and across VQC, compositional, quantum-transformer, and quantum-kernel model families alike, with no pairing standing out as a consistent winner once the small per-cell counts are taken into account. This is not surprising in hindsight: encoding choice inside these two task families tracks the corpus-wide habits documented in Sections V and VI, mostly which toolchain and which pretrained embedding a group already had on hand, rather than anything about sentiment or topic classification specifically. Because dozens of unrelated research groups are all applying broadly interchangeable recipes to broadly interchangeable binary and multi-class labeling problems, classification is where the corpus has the most data and the least power to say anything about task-encoding fit.

Similarity and disambiguation is the one task family where a real, mechanistically grounded pattern emerges. Amplitude encoding accounts for close to half of these studies, roughly double its share of the corpus overall. This is not a correlation in search of an explanation. Section V-B already established why: an amplitude-encoded state's inner product literally is a cosine similarity, so a SWAP test or overlap measurement performs the task's core computation rather than merely feeding a downstream classifier. The same mechanism explains why these studies cluster in the quantum-kernel model family: a kernel method is little more than a trained readout wrapped around exactly that overlap. Claimed advantage in this cluster is also more consistent than elsewhere, as one would expect when an encoding's mathematical structure matches the task rather than merely hosting it. This is, in effect, the one place in the corpus where the encoding does the work the task actually needs. It is telling that the field found this fit by mathematical necessity rather than by systematic search: no study set out to compare encodings on similarity tasks and arrived at this conclusion. It falls out of what amplitude encoding is.

Relational tasks—question answering and, to a lesser extent, similarity itself—show a weaker echo of the same logic on the compositional side. Syntactic/DisCoCat encoding is disproportionately present here relative to its footprint in classification, which fits the intuition that grammatical structure should matter most when a task is explicitly about how words or entities relate. The claimed-advantage picture in this corner, however, is considerably more hedged: mostly conditional rather than an unqualified yes. That hedging is consistent with, not contrary to, Section V-C's finding that the family's practicality erodes precisely as a task departs from a short, curated sentence. Relational tasks are exactly where compositional structure is best motivated and exactly where the grammar-to-circuit pipeline is under the most strain. The two effects partly cancel, and the result is theoretical fit without a reliably delivered edge.

Sequence labeling, question answering, and generation or language modeling are each covered by only six to ten studies, too few to support a claim either way, but they fail differently. Sequence labeling is almost entirely angle-encoded, so there is nothing to compare, one encoding occupies the cell rather than competing against alternatives inside it. Question answering and generation, by contrast, are spread thinly across several encodings, but each individual pairing is one or two papers from a single research thread rather than a pattern a second, independent group has reproduced. Neither failure mode supports a task-encoding claim, but for opposite reasons: one cell has no variation to explain, the others have too little replication to trust. This section's central finding is therefore not a list of winning task-encoding pairings but a methodological one. The advantage label is self-reported by each paper's own authors against whatever classical baseline and dataset they individually chose, not computed against a shared benchmark. A raw difference in claimed-advantage rates between encodings is therefore weak evidence on its own. It becomes trustworthy only when, as with amplitude encoding on similarity tasks, it coincides with an independent structural reason the encoding should help. On the present corpus, that coincidence happens exactly once.

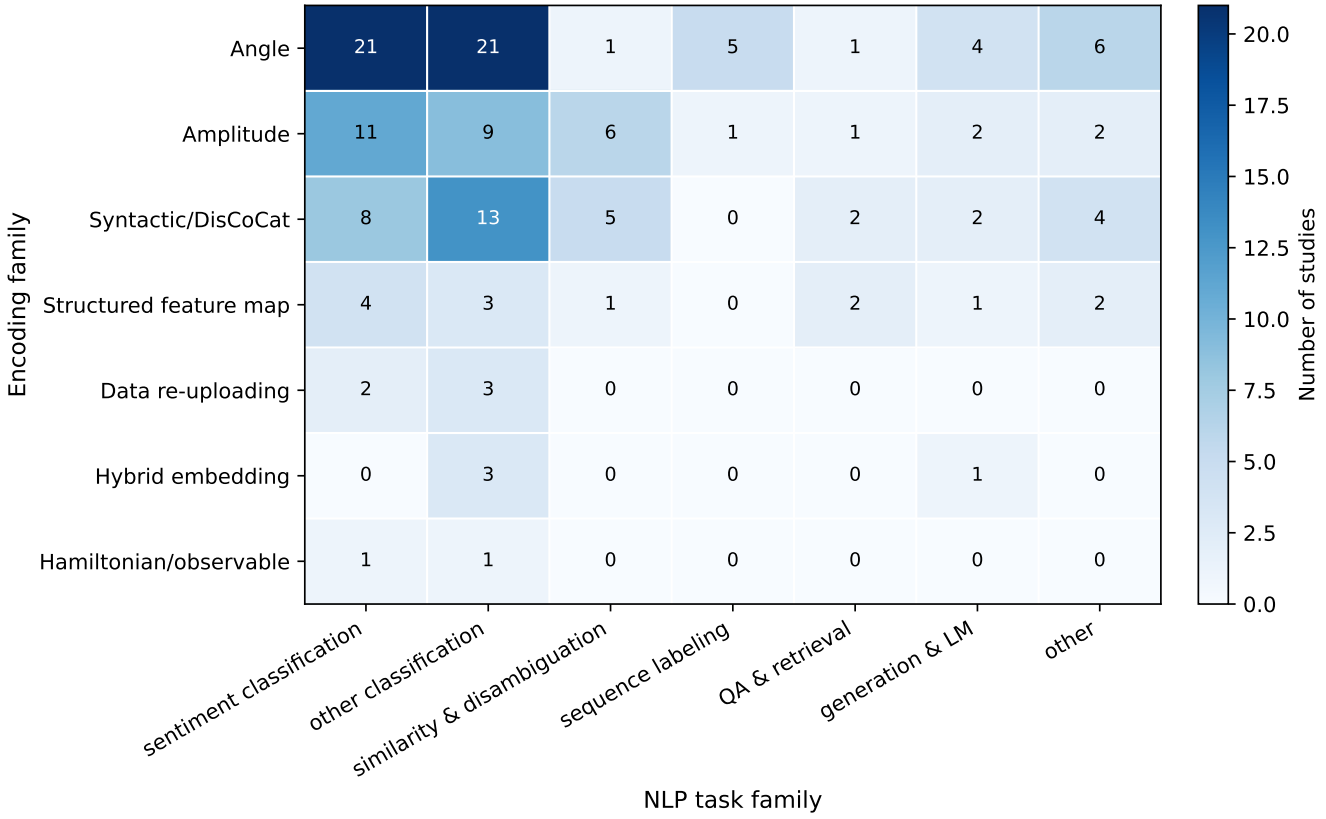



**FIGURE 5.** Encoding family by NLP task family in the final primary corpus. A study evaluating several task families is counted once in each.

## VIII. EXPERIMENTAL RIGOR: SIMULATION, HARDWARE, AND REPRODUCIBILITY

The preceding sections asked what is encoded and by which model; this section asks how seriously the corpus tests any of it. Every primary study was coded for its execution regime, exact noiseless statevector simulation, simulation under an explicit noise model, execution on real quantum hardware, or a mixed protocol combining simulation with a hardware run, together with the specific framework and backend used, whether code was released, and whether trainability (barren plateaus) and scalability were discussed. These fields describe the experimental apparatus underneath the encoding and model-family results of Sections V–VII and let us ask a question those sections cannot: not which encoding is used, but how much evidence backs the claim that it works.

### A. EXECUTION REGIMES ACROSS THE CORPUS

Noiseless statevector simulation dominates: 95 of 125 studies (76.0%) report exact simulation only. A further 14 (11.2%) simulate under an explicit noise model, typically depolarizing or readout error injected into a density-matrix simulator, and 14 (11.2%) report a mixed protocol that pairs simulation with a validation run on real hardware. Only two studies (1.6%) run exclusively on real hardware with no simulated baseline reported: a quantum-kernel KNN classifier executed on IBM Quantum Experience [46] and a quantum self-attention model executed on the 72-qubit Wukong superconducting processor using a 28-qubit register [101]. Combining the mixed and hardware-only categories, 26 studies (20.8%) touch real hardware in some capacity; the remaining four-fifths of the corpus never leaves a simulator.

**TABLE 9.** Execution regime by encoding family in the final primary corpus. "HW-touched" sums the mixed and real-hardware columns.

| Encoding family | Noiseless | Noisy sim | Mixed | HW-only | HW-touched % |
|---|---|---|---|---|---|
| Angle | 42 | 6 | 2 | 1 | 17.6 |
| Amplitude | 23 | 2 | 2 | 1 | 17.9 |
| Syntactic/DisCoCat | 19 | 4 | 6 | 0 | 34.5 |
| Structured feature map | 5 | 1 | 3 | 0 | 44.4 |
| Data re-uploading | 2 | 1 | 1 | 0 | 50.0 |
| Hybrid embedding | 3 | 0 | 0 | 0 | 0.0 |
| Hamiltonian/observable | 1 | 0 | 0 | 0 | 0.0 |

Table 9 shows that hardware engagement is not evenly spread across the taxonomy of Section IV: it concentrates in exactly the families that Section V already identified as small and toolchain-bound. Syntactic/DisCoCat encoding, at 34.5%, is validated on real devices far more often than the corpus average, because the lambeq/DisCoPy/pytket stack that dominates the family (Section V-C) ships with direct IBM Quantum integration, and the shallow, few-qubit circuits that a short sentence produces are cheap enough to queue

on real devices: reported backends include `ibmq_bogota` [18], `ibmq_montreal` and `ibmq_toronto` [20], `ibm_lagos`/`ibm_oslo`/`ibm_nairobi` [57], and the Quantinuum trapped-ion device [138]. Structured feature maps and data re-uploading show the highest hardware-touched shares of all (44.4% and 50.0%), but on denominators of nine and four studies respectively, so these percentages describe two or three papers each rather than a stable pattern; the corpus is too thin here to call the concentration a trend. Angle and amplitude encoding, which together account for 79 of the 125 studies and for almost every VQC- and recurrent-model result discussed in Section V-A and Section V-B, sit at the bottom of the table: barely one study in six ever leaves the simulator. This means the two encodings that carry the field's growth (Section VI-A) are also the two for which almost all reported accuracy numbers are simulator artifacts, since a noiseless statevector simulation, unlike a hardware run or even a calibrated noisy simulation, is silent on whether the same circuit would survive decoherence, gate error, or readout noise on an actual device.

Hybrid embedding and Hamiltonian/observable encoding report no hardware or noisy-simulation validation whatsoever: every one of these studies is exact-simulation only. For Hamiltonian encoding this is a single study, so no pattern can be claimed, but it does mean the family's central promise, that observable-based readout sidesteps the state-preparation cost that undermines amplitude encoding (Section V-E), has not yet been tested under any of the conditions, noise or hardware, that would actually stress that claim.

### B. EXECUTION BY MODEL FAMILY AND THE RECURRENT-NETWORK BLIND SPOT

The same asymmetry appears when execution is crossed with model family instead of encoding. Quantum kernels are the most hardware-engaged family (5 of 13 studies, 38.5%, touch hardware or a noisy simulator), consistent with their concentration in amplitude and structured-feature-map encoding (Section V-B); compositional models follow at 34.5%, the same figure as syntactic encoding, since the two categories are the same 29 studies (Section V-C). VQCs, hybrid-neural models, and quantum transformers cluster around 19–26% hardware or noisy-simulation engagement. Recurrent quantum models are the one family with zero: all 12 recurrent-network studies in the corpus (QLSTM, QRNN, and related sequential architectures) report noiseless simulation only, with no noisy-simulation or hardware run reported for any of them. This is worth flagging on its own terms, independent of the encoding-level pattern in Section VIII-A, because a recurrent quantum cell re-applies its encoding and ansatz once per token (Section V-A), so circuit depth, and with it the exposure to decoherence and gate error, scales with sequence length in a way that a single-shot VQC does not. The recurrent family is therefore the one whose central architectural claim, that a small register can be reused cheaply across a sequence, is also the one for which noise robustness has been tested least, not most.

### C. THE HARDWARE-VALIDATION TIMELINE IS FLATTENING, NOT GROWING

Section VI-A showed that the corpus grew roughly twenty-fold in raw study count between 2021 and 2025. The hardware-touched share (mixed and real-hardware execution combined) does not grow with it. It rises from 0% in 2021 to 37.5% in 2022 (3 of 8 studies) and stays in the 14–35% range through 2024 (28.6% in 2023, 34.6% in 2024, the latter also the only year with a hardware-only study), then falls to 2.4% in 2025 (1 of 41 studies) and 3.7% in 2026 (1 of 27, partial year). Noisy-simulation share follows a milder version of the same late dip, 14–15% in 2023, 2025, and 2026 but only 3.8% in 2024. Read together, the two most recent full or near-full years in the corpus, 2025 and 2026, are the ones with the lowest combined rate of hardware or noise engagement of any period after 2021, at the exact moment the field's yearly output peaked. This is the experimental-rigor counterpart to the mechanism Section VI-A already proposed for the encoding mix: if the field's growth ran through the encodings and model families that asked the least of a newcomer, VQCs and hybrid-neural models consuming pretrained embeddings rather than DisCoCat's grammar pipeline, then it plausibly also ran through the experimental protocol that asks the least of a newcomer, a local noiseless simulator call, rather than a queued and rate-limited hardware job or a calibrated noise model. The one exception the timeline makes room for is the 2026 quantum-secure federated-learning study, which fine-tunes a hybrid classical–quantum sentiment model on a four-client quantum network and validates it "on real quantum hardware" over the 156-qubit BAQIS Quafu cloud backend, alongside large-scale classical simulation for scalability [143]; it is the corpus's only 2025–2026 hardware-touched NLP study and, notably, is motivated by a quantum-communication security goal rather than by an NLP-performance claim, which is why it does not read as a counter-trend so much as a study that reached hardware for reasons orthogonal to the pattern above.

### D. FRAMEWORK FRAGMENTATION AND THE REPRODUCIBILITY GAP

Even where a study stays in simulation, the tooling underneath it is far from standardized. Across the corpus, 34 of 125 studies (27.2%) do not report the simulation framework at all, the single largest category, larger than any one named framework. Among studies that do report one, PennyLane is the most common (21 as sole framework, plus another dozen or so in combination with Qiskit, PyTorch, or a hardware backend), Qiskit is second (9 sole, plus combinations), and lambeq anchors the syntactic/DisCoCat subcorpus as documented in Section V-C, but the remaining reported configurations are almost all unique to a single paper, TensorCircuit with IBM-calibration-derived depolarizing noise [108], Amazon Braket Hybrid Jobs, Qulacs, TensorFlow Quantum, Flax, so that no two studies outside the lambeq cluster and a PennyLane/Qiskit core are running directly comparable software stacks. This fragmentation compounds

a second gap: only 25 of 125 studies (20.0%) report that code is publicly available. Between the missing-framework rate and the missing-code rate, a substantial fraction of the corpus, plausibly overlapping but not identical sets, cannot be independently re-run even in simulation, let alone re-validated on hardware.

Reporting on trainability is thinner still. Barren plateaus, the vanishing-gradient pathology that is a known risk for exactly the wide, deep, or highly entangling circuits that Section V-A and Section V-B describe (28-qubit attention registers [101], ten-qubit-plus amplitude loads [43], [44]), is substantively discussed in only 4 studies (3.2%); a further 19 (15.2%) mention the term without engaging it, and 102 (81.6%) do not raise it at all. Scalability fares better on paper, 91 studies (72.8%) are coded as discussing scalability against 34 that do not, but this field, like the quantum-advantage label examined in Section VII, is self-reported by each paper against whatever standard its own authors chose; a scalability discussion here can range from a formal resource analysis to a single sentence noting that larger qubit counts are future work. Taken together with the noiseless-simulation dominance of Section VIII-A, the corpus's own reporting practice under-tests the two properties, noise robustness and trainability at scale, that determine whether any encoding surveyed in Section IV would still work outside a simulator, which is the same caution Section IV-B already raised for amplitude encoding's hidden state-preparation cost, now shown to generalize across the corpus rather than being specific to one family.

## IX. TRENDS AND OPEN CHALLENGES

The preceding sections examined the corpus from five angles in turn: what is encoded (Section IV), how the same encoding is repurposed across model families (Section V), how the field's composition changed over time (Section VI), whether the choice ever tracks the task (Section VII), and how much experimental evidence actually backs any of it (Section VIII). This section pulls those five threads together into a single account of where the field is heading, states plainly what the encoding bottleneck named in the abstract actually is, and converts the gaps documented above into a concrete agenda for future work.

### A. WHERE THE FIELD IS HEADING

Table 10 first fixes the quantitative baseline: for each categorical dimension of the comparison framework (Section III-F), it reports the distribution over the full corpus of 125 studies. Three trends, each established separately above, describe the same underlying movement when read together.

First, the field is consolidating around encodings that make the fewest demands on a newcomer: angle and amplitude encoding, which accept any fixed-length numeric vector and therefore compose freely with whatever pretrained embedding a research group already has on hand, now account for four of every five studies published from 2024 onward (Section VI-A), while syntactic/DisCoCat encoding, which requires a grammar formalism and a narrow parser toolchain, has fallen from the entirety of the 2021 corpus to roughly one study in nine by 2026 (Section VI-B). Second, this consolidation is a consolidation onto simulators, not onto quantum hardware: the years in which the corpus grew fastest, 2025 and 2026, are also the years with the lowest combined rate of hardware or noisy-simulation engagement of the entire six-year window (Section VIII-C). Third, where the corpus does show a real, mechanistically grounded success, amplitude encoding on similarity and disambiguation tasks, where the encoded state's inner product literally performs the task's core computation (Section VII), that success is small, concentrated in a single task family and a single model family (quantum kernels), and was arrived at by mathematical necessity rather than by systematic search across the corpus. Put together, the field's growth curve and its evidentiary curve are decoupling: publication volume is rising fastest in exactly the segment, simulator-only VQC and hybrid-neural pipelines built on commodity embeddings, that is least equipped to demonstrate anything about noise, hardware behavior, or a genuine quantum contribution, while the segment with the clearest structural argument for a quantum-native advantage remains a small, slow-growing niche.

**TABLE 10.** Distribution of the framework's categorical dimensions over the final primary corpus ($n = 125$). All counts are computed directly from the released extraction sheet.

| Dimension | Value | $n$ | % |
|---|---|---|---|
| Execution regime | noiseless simulation | 95 | 76.0 |
| | noisy simulation | 14 | 11.2 |
| | mixed (sim. + hardware) | 14 | 11.2 |
| | real hardware only | 2 | 1.6 |
| Advantage claim | conditional | 46 | 36.8 |
| | yes | 41 | 32.8 |
| | no | 38 | 30.4 |
| Encoding bottleneck | implicit | 55 | 44.0 |
| | absent | 36 | 28.8 |
| | explicit | 34 | 27.2 |
| Barren plateaus | absent | 102 | 81.6 |
| | mentioned | 19 | 15.2 |
| | discussed | 4 | 3.2 |
| Qubit scaling tested | no | 102 | 81.6 |
| | yes | 23 | 18.4 |
| Scalability discussed | yes | 91 | 72.8 |
| | no | 34 | 27.2 |
| Classical baseline | yes | 110 | 88.0 |
| | no | 15 | 12.0 |
| Code available | no | 100 | 80.0 |
| | yes | 25 | 20.0 |

### B. THE ENCODING BOTTLENECK, STATED FORMALLY

Every family reviewed in Sections IV and V is, underneath its individual mechanism, a different response to the same arithmetic. Let $d$ denote the dimensionality of the text representation a study starts from, ranging in this corpus from a handful of TF-IDF or handcrafted features up to the 768 or more dimensions of a transformer embedding, and let $q$ denote the near-term qubit budget a study actually uses, empirically two to sixteen qubits across the angle and amplitude subcorpora (Tables 4 and 5). Angle encoding maps at most one feature to one rotation, so it requires $q \geq d'$ where $d' \leq d$ is whatever survives a classical reduction

step applied before the circuit is built (Section IV-A); the mismatch between $d$ and $q$ is therefore resolved by discarding information, and the quantum circuit only ever sees the $d'$-dimensional remainder. Amplitude encoding instead packs the full vector into $q = \lceil \log_2 d \rceil$ qubits, so the register grows logarithmically rather than linearly with $d$, but preparing an arbitrary $2^q$-dimensional amplitude vector generically requires a state-preparation circuit whose depth scales with $d$ itself (Section IV-B); the mismatch is resolved by paying in circuit depth instead of in discarded information, a cost that Section V-B showed is routinely absorbed into a simulator call and left out of the reported qubit budget entirely. Structured feature maps and data re-uploading sit between these two extremes, trading some width for some depth without eliminating either cost (Sections IV-D and V-E), and hybrid embedding relocates the mismatch upstream into a classical encoder rather than resolving it. Syntactic/DisCoCat encoding is the one family that sidesteps the arithmetic rather than trading inside it, because $q$ is set by the grammatical structure of the sentence rather than by the dimensionality of an embedding (Section IV-C), which is exactly why Section IV treats it as categorically different from the other five families and why it is the one encoding whose qubit count Section V-C found to be untouched by ansatz choice.

This is the encoding bottleneck named in the abstract: a persistent gap between $d$, set by what classical NLP representations require to be informative, and $q$, set by what near-term hardware can support, such that every non-syntactic encoding family in the corpus is best understood as a choice of which cost absorbs the gap, lost information, added depth, or a relocated classical burden, rather than a choice that closes the gap itself. No study in the corpus reports an encoding that grows the effective qubit budget without incurring one of these three costs, which is the sense in which the bottleneck is structural rather than a matter of engineering effort within any single family.

### C. THE RIGOR GAP: WHAT GROWTH HAS NOT SOLVED

Section VIII documented the pieces of this gap individually; read as a whole, they describe a field whose reporting practice has not kept pace with its output. Three-quarters of the corpus never leaves a noiseless simulator, and the two families carrying the field's growth, angle and amplitude encoding, are the least hardware-validated of all (Section VIII-A). Barren plateaus are substantively discussed in one study in thirty despite being a known risk for the wide or deep circuits several of the surveyed model families construct (Section VIII-D). More than a quarter of the corpus does not report which simulation framework was used, and four in five studies release no code, so a meaningful share of the corpus cannot be independently reproduced even in simulation, let alone re-validated on hardware. Underneath all of this sits the finding of Section VII: the quantum-advantage label itself is self-reported by each paper against whatever classical baseline, dataset, and metric its own authors chose, so raw counts of claimed advantage carry little evidential weight on their own and become trustworthy only when, as with amplitude encoding on similarity tasks, they coincide with an independent structural reason the encoding should help. None of these gaps is specific to one encoding family or one model family; they are properties of how the field as a whole reports its experiments, which is why closing them is a precondition for the task-encoding fit questions raised in Section VII to be answered with any confidence beyond the single case the current corpus supports.

### D. OPEN CHALLENGES

The gaps documented throughout the review reduce to seven open challenges, each derived from the corpus itself rather than imported from general quantum-computing commentary. For each, we state the challenge, the corpus evidence with counts from Table 10, and why it remains open; Table 11 summarizes them.

C1) The encoding bottleneck itself. State-preparation and compression costs dominate any downstream benefit. Evidence: 89 of 125 studies (34 explicit, 55 implicit) shape their design around the $d$-versus-$q$ mismatch formalized in Section IX-B; the corpus's responses, divide-and-conquer preparation [49], observable encoding [77], depth-for-width trades [72], mitigate the cost without closing the gap. Open because no surveyed encoding grows the effective qubit budget without paying in information, depth, or a relocated classical burden.

C2) Non-demonstrable quantum advantage. The advantage label is self-reported against author-chosen baselines. Evidence: 41 studies claim advantage, 46 conditionally, 38 not at all, while 15 report no classical baseline whatsoever; the one cluster where claims align with a structural reason is amplitude encoding on similarity tasks [46], [51]. Open because no shared benchmark with fixed classical baselines exists against which the claims could be made commensurable.

C3) Trainability at scale (barren plateaus). Evidence: 102 studies never raise barren plateaus, 19 mention the term without engaging it, and only 4 analyze it, despite corpus registers reaching 28 qubits [101] and only isolated ansatz-level analyses [53]. Open because the corpus's dominant circuit sizes are exactly where the pathology is mild, so present results say little about the widths any scaling path requires.

C4) The simulation–hardware gap. Evidence: 95 studies (76.0%) never leave a noiseless simulator; hardware-touched work peaked in 2022–2024 and fell to 2.4% and 3.7% of the 2025 and 2026 corpora respectively (Section VIII-C), with [143] the only recent hardware-validated study. Open because noiseless simulation is silent on decoherence, gate error, and readout noise, precisely the properties that decide near-term viability.

C5) Absent scaling evidence. Evidence: 102 of 125 studies evaluate a single register width; only 23 test more than one, and systematic sweeps such as the 1–16-

**TABLE 11.** Open challenges with corpus evidence. Counts are over the full corpus ($n = 125$) and computed from the released extraction sheet.

| Challenge | Corpus evidence | Representative studies |
|---|---|---|
| C1 Encoding bottleneck | 89/125 designs shaped by it (34 explicit, 55 implicit) | [49], [72], [77] |
| C2 Advantage non-demonstrable | 41 yes / 46 conditional / 38 no; 15 without classical baseline | [46], [51] |
| C3 Barren plateaus untreated | 102/125 silent; 4 substantive analyses | [53], [101] |
| C4 Simulation–hardware gap | 95/125 noiseless only; 2025–26 hardware share $\leq$3.7% | [101], [143] |
| C5 No scaling evidence | 102/125 single register width | [49], [73] |
| C6 Reproducibility & benchmarks | 100/125 no code; 34/125 framework unreported | [18], [56], [134] |
| C7 Structure lost in compression | 768-dim embeddings cut to 2–10 features; 34/125 no scalability discussion | [36], [37], [79] |

qubit comparison of [49] are rare. Open because every resource argument in Section IX-B is asymptotic while the evidence is single-point.

C6) Reproducibility and benchmark triviality. Evidence: 100 studies (80.0%) release no code and 34 (27.2%) do not name their simulation framework (Section VIII-D); much of the compositional subcorpus evaluates on the $\sim$100-sentence MC/RP datasets [18], [134], where perfect accuracies are reported on small samples [56]. Open because neither independent re-execution nor cross-paper comparison is possible at this reporting standard.

C7) Loss of linguistic structure under compression. Evidence: outside the syntactic family, the reduction stage discards most of the representation before the circuit sees it, 768-dimensional embeddings reduced to ten [36], four [37], or two [79] features, and 34 studies do not discuss scalability of this pipeline at all. Open because no surveyed study quantifies what the reduction removes, so the quantum circuit's input is uncharacterized in exactly the dimension the field claims to process.

### E. AN AGENDA FOR FUTURE WORK

Four actionable gaps follow directly from challenges C1–C7 above.

Noise-in-the-loop evaluation as a default, not an afterthought. Because angle and amplitude encoding are simultaneously the fastest-growing families (Section VI-A) and the least hardware- or noise-validated (Section VIII-A), and because recurrent quantum architectures re-apply their encoding once per token with no noisy-simulation or hardware study reported anywhere in the corpus (Section VIII-B), the highest-value next step is not more noiseless benchmarking but systematic reporting of accuracy under an explicit, calibrated noise model as a minimum bar for publication, following the small set of studies in the corpus that already model depolarizing and readout error from real device calibration data rather than treating noise as a qualitative caveat.

Controlled, cross-family encoding comparisons. Section V showed that encoding and model family are so tightly coupled in the corpus that comparing "encoding families" is often really comparing the architectures they are bound to; the small number of studies that hold model family, dataset, and circuit depth fixed while swapping only the encoding or ansatz (Sections V-C and V-D) are disproportionately informative relative to their count and should be the template, not the exception.

Shared benchmarks for the quantum-advantage claim. Since Section VII found the advantage label uninterpretable across studies precisely because each paper chooses its own classical baseline and dataset, a small set of shared, encoding-agnostic NLP benchmarks with fixed classical baselines, of the kind proposed piecemeal but not standardized in the surveyed studies, would let future meta-analyses attribute a performance gap to the encoding rather than to which baseline an individual group happened to select.

Extending amplitude encoding's structural fit beyond similarity tasks. The one place in the corpus where an encoding's mathematical structure matches the task rather than merely hosting it, amplitude encoding on similarity and disambiguation, was found by mathematical necessity rather than deliberate search (Section VII). Whether an analogous structural fit exists for other task families, for instance whether the compositional structure of syntactic encoding could be matched more deliberately to relational tasks such as question answering, where Section VII found a theoretical but unreliably delivered edge, is an open question the current corpus is too thin to answer but that a targeted study, rather than another general-purpose classifier, could address directly.

## X. CONCLUSION

This survey set out from a simple observation: every QNLP system must, at some point, convert text into a quantum state or circuit parameter, yet this step is almost never treated as a variable in its own right. Most primary studies name an encoding in passing and move directly to reporting accuracy, leaving no basis for comparing what different encodings actually cost or preserve. To close that gap, this article conducted a pre-registered systematic literature review under a protocol fixed before the search was executed (Section III), screened 376 database records and 1222 snowball candidates down to a final corpus of 125 primary studies published between 2021 and 2026, and built a six-family taxonomy, angle, amplitude, syntactic/DisCoCat, structured feature-map, data re-uploading, and hybrid embedding, organized not by model architecture or ansatz but by the point at which text enters the quantum computation (Section IV).

Four findings recur across the analysis. First, encoding and model family are tightly coupled rather than independently chosen: syntactic encoding and compositional QNLP are the same 29 studies, structured feature maps concentrate in quantum kernels, and only angle and amplitude encoding travel freely across architectures, which means that compar-

ing "encodings" in the literature is frequently a comparison of the architectures they are bound to rather than of the encoding step in isolation (Section V). Second, the field's rapid growth since 2023 has been a consolidation onto the encodings and model families that ask the least of a newcomer, angle- and amplitude-encoded VQCs and hybrid-neural pipelines built on pretrained embeddings, at the expense of the grammar-dependent compositional tradition that defined the field's first two years (Section VI). Third, that same growth has not been matched by growth in experimental rigor: three-quarters of the corpus never leaves a noiseless simulator, hardware and noisy-simulation engagement peaked in 2022–2024 and fell to its lowest levels in the very years the corpus grew fastest, and barren plateaus, code availability, and simulation framework are each under-reported by a wide margin (Section VIII). Fourth, claimed quantum advantage is self-reported against whatever baseline each paper's own authors chose, which makes it weak evidence on its own; it becomes trustworthy only in the one case where it coincides with a structural reason the encoding should help, amplitude encoding on similarity and disambiguation tasks, where the encoded state's inner product performs the task's core computation directly (Section VII).

Underlying all four findings is the encoding bottleneck formalized in Section IX-B: the persistent gap between the dimensionality of a useful text representation and the qubit budget near-term hardware can support. Every encoding family surveyed, syntactic encoding excepted, resolves this gap by paying in one of three currencies, discarded information, added circuit depth, or a relocated classical burden, and none eliminates it. Framing the encoding step this way is the taxonomy's main payoff: it turns a fragmented set of implementation choices into a small number of comparable trade-offs, and it is the basis on which Section IX converts the review's gaps, untested noise robustness, uncontrolled cross-family comparisons, non-standardized advantage claims, and an underexplored task-encoding fit beyond similarity, into a concrete agenda for the studies that follow.

This review has its own limits, which should condition how its conclusions are used. The search covered two databases, Scopus and IEEE Xplore, and while two rounds of backward and forward snowball sampling were used specifically to recover work those databases miss, a search strategy of this kind cannot guarantee complete coverage of a fast-moving, preprint-heavy field. The 2026 publication counts used throughout are necessarily partial, since the search was conducted before the year closed, and every year-over-year trend reported in Sections VI and VIII should be read with that caveat attached to its final data point. Several of the extracted fields, quantum-advantage claims, scalability discussion, and stated limitations among them, are self-reported by the primary studies themselves rather than independently re-derived, so the review inherits the same inconsistency in how individual authors define and report these properties that Section VII and Section VIII-D identify as a limitation of the field at large.

None of this changes the central conclusion. Text encoding is not a neutral preprocessing step in QNLP; it is the decision that fixes how much of a language sample survives contact with a near-term quantum circuit, and it deserves the same scrutiny as the ansatz, the optimizer, or the classical baseline it is compared against. The taxonomy, corpus, and extracted metadata released with this survey are offered as the basis for that scrutiny in future work, including the encoding-controlled comparisons, noise-in-the-loop evaluations, and shared benchmarks called for in Section IX.

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## APPENDIX A CORPUS CLASSIFICATION

Tables 12–14 list the complete classification of the 125 primary studies under the encoding comparison framework of Section III-F: model family, encoding family, qubit count, primary dataset, and hardware backend for each study. The tables are generated directly from the released extraction sheet, so they can be regenerated whenever the corpus data is revised.

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**TABLE 12.** Classification of the final primary corpus under the encoding comparison framework (part 1 of 3): study, model family, encoding family, qubit count, primary dataset, and hardware backend. Generated from the released extraction sheet by scripts/generate_slr_tables.py.

| Study | Model family | Encoding | Qubits | Primary dataset | Backend |
|---|---|---|---|---|---|
| [19] | compositional | syntactic | NR | NR | NA |
| [126] | compositional | syntactic | 2 | Meaning Classification (Lorenz | NA |
| [54] | compositional | syntactic | NR | restaurant sentiment dataset | NA |
| [129] | compositional | syntactic | 7 | custom intermediate dataset | IBM Qiskit Aer simulat |
| [135] | compositional | syntactic | NR | pronoun resolution dataset (14 | NA |
| [138] | compositional | syntactic | 25 | Custom musical compositions da | Quantinuum |
| [55] | compositional | syntactic | NR | Kinetics (subset) | NA |
| [136] | compositional | syntactic | NR | custom Chinese sentence corpus | NA |
| [59] | compositional | syntactic | 5 | custom restaurant domain sente | NA |
| [20] | compositional | syntactic | 2 | custom relative pronoun corpus | ibmq_montreal |
| [18] | compositional | syntactic | 7 | MC | ibmq_bogota |
| [140] | compositional | syntactic | NR | custom topic classification da | NA |
| [134] | compositional | syntactic | NR | MC | Qiskit Aer simulator |
| [132] | compositional | syntactic | 3 | custom simple sentence classif | IBM Quantum Experience |
| [56] | compositional | syntactic | NR | NR | NA |
| [133] | compositional | syntactic | NR | restaurant domain sentences (s | simulated noisy model |
| [141] | compositional | syntactic | NR | NR | NA |
| [57] | compositional | syntactic | 7 | synthetic datasets | ibm_lagos |
| [58] | compositional | syntactic | NR | custom Winograd-inspired prono | NA |
| [127] | compositional | syntactic | 12 | compositional image captioning | NA |
| [142] | compositional | syntactic | 20 | IMDB database (SQL queries) | NA |
| [128] | compositional | syntactic | NR | NR | Qiskit Aer simulator |
| [60] | compositional | syntactic | 10 | 50 hand-crafted sentences (sim | Qiskit statevector sim |
| [53] | compositional | syntactic | 4 | custom restaurant domain sente | Qiskit Aer Simulator |
| [139] | compositional | syntactic | 10 | STS | NA |
| [4] | compositional | syntactic | NR | mc data (IT vs food topic) | NA |
| [137] | compositional | syntactic | 13 | English-Persian sentence pairs | IBMQ qasm simulator |
| [131] | compositional | syntactic | 11 | SICK | NA |
| [130] | compositional | syntactic | NR | manually curated Hindi sentenc | NA |
| [124] | hybrid_nn | amplitude | NR | Prolog benchmark | NA |
| [122] | hybrid_nn | amplitude | 6 | WN18RR | NA |
| [123] | hybrid_nn | amplitude | 12 | real-world multimodal sentimen | NA |
| [28] | hybrid_nn | angle | 6 | Mustard | NA |
| [93] | hybrid_nn | angle | NR | The Black Swan generated stega | NA |
| [89] | hybrid_nn | angle | NR | The Gift of the Magi (O. Henry | NA |
| [95] | hybrid_nn | angle | 5 | BOOKCORPUS | noisy quantum simulati |
| [94] | hybrid_nn | angle | NR | synthetic (pattern reproductio | NA |
| [90] | hybrid_nn | angle | 4 | Coronavirus Tweets Dataset (CV | NA |
| [92] | hybrid_nn | angle | 2 | MELD | NA |
| [91] | hybrid_nn | angle | NR | CMU-MOSI | PennyLane lightning-gp |
| [35] | hybrid_nn | angle | 4 | synthetic data | NA |
| [38] | hybrid_nn | angle | 8 | ASTD (Arabic Sentiment Twitter | Qiskit Aer simulator |

**TABLE 13.** Classification of the final primary corpus under the encoding comparison framework (part 2 of 3): study, model family, encoding family, qubit count, primary dataset, and hardware backend. Generated from the released extraction sheet by scripts/generate_slr_tables.py.

| Study | Model family | Encoding | Qubits | Primary dataset | Backend |
|---|---|---|---|---|---|
| [76] | hybrid_nn | hybrid_embed | NR | FakeNewsNet | NA |
| [62] | hybrid_nn | structured_feature_map | 5 | SST-2 | IBM Yonsei |
| [116] | quantum_kernel | amplitude | 7 | movie reviews (custom 50-examp | IonQ simulator |
| [115] | quantum_kernel | amplitude | 10 | fastText | NA |
| [46] | quantum_kernel | amplitude | 11 | SMS Spam Collection | IBM Quantum Experience |
| [51] | quantum_kernel | amplitude | NR | word analogy datasets | NA |
| [45] | quantum_kernel | amplitude | 12 | SimLex-999 (666 noun pairs) | NA |
| [107] | quantum_kernel | angle | 4 | POS tagging benchmark (dataset | NA |
| [31] | quantum_kernel | angle | NR | UCI ML Drug Review | NA |
| [68] | quantum_kernel | structured_feature_map | 6 | Bengali news sentiment | IBM noisy qasm_simula |
| [65] | quantum_kernel | structured_feature_map | 14 | restaurant domain sentences (2 | adhoc Qiskit quantum k |
| [61] | quantum_kernel | structured_feature_map | 20 | custom word-topic datasets | IonQ Aria |
| [63] | quantum_kernel | structured_feature_map | NR | SMS Spam Collection | ibm_osaka |
| [64] | quantum_kernel | structured_feature_map | 10 | HARD Arabic hotel reviews | PennyLane Lightning ex |
| [67] | quantum_kernel | structured_feature_map | NR | Wikispeedia | NA |
| [117] | quantum_transformer | amplitude | 4 | NR | NA |
| [118] | quantum_transformer | amplitude | 5 | NR | NA |
| [119] | quantum_transformer | amplitude | 10 | Yelp | NA |
| [121] | quantum_transformer | amplitude | 12 | SemCor 3.0 | NA |
| [34] | quantum_transformer | angle | 4 | IMDB | NA |
| [26] | quantum_transformer | angle | NR | ColBERT | NA |
| [41] | quantum_transformer | angle | 4 | VQA v2 | Qiskit Aer IBM stateve |
| [102] | quantum_transformer | angle | 4 | SST-2 | NA |
| [101] | quantum_transformer | angle | 28 | MC (Meaning Classification) | Wukong (72-qubit super |
| [104] | quantum_transformer | angle | NR | NR | NA |
| [103] | quantum_transformer | angle | 4 | Yelp | NA |
| [105] | quantum_transformer | angle | NR | MVSA-Single | NA |
| [106] | quantum_transformer | angle | 5 | Yelp | NA |
| [69] | quantum_transformer | structured_feature_map | NR | ATCOSIM | NA |
| [125] | recurrent_nn | amplitude | NR | Chinese sentiment dataset | NA |
| [29] | recurrent_nn | angle | 4 | ChatGPT-generated financial se | NA |
| [97] | recurrent_nn | angle | NR | code-mixed social media | NA |
| [39] | recurrent_nn | angle | 4 | Bengali text classification da | NA |
| [25] | recurrent_nn | angle | NR | English-Spanish parallel corpu | NA |
| [100] | recurrent_nn | angle | 4 | Rotten Tomatoes | Quantinuum emulator |
| [24] | recurrent_nn | angle | 8 | Indiana University Chest X-ray | PennyLane quantum simu |
| [96] | recurrent_nn | angle | NR | 9 code-mixed datasets | NA |
| [98] | recurrent_nn | angle | 5 | Sentiment140 | NA |
| [30] | recurrent_nn | angle | 8 | MIMIC-III discharge summaries | NA |
| [99] | recurrent_nn | angle | 4 | Rotten Tomatoes | NA |
| [71] | recurrent_nn | data_reuploading | NR | NVD | NA |
| [43] | vqc | amplitude | 10 | ItaCoLa | NA |

**TABLE 14.** Classification of the final primary corpus under the encoding comparison framework (part 3 of 3): study, model family, encoding family, qubit count, primary dataset, and hardware backend. Generated from the released extraction sheet by scripts/generate_slr_tables.py.

| Study | Model family | Encoding | Qubits | Primary dataset | Backend |
|---|---|---|---|---|---|
| [49] | vqc | amplitude | 16 | lambeq (MC) | ibm_kyoto |
| [111] | vqc | amplitude | NR | code-mixed social media datase | NA |
| [110] | vqc | amplitude | 10 | SentiPolc Italian sentiment da | NA |
| [112] | vqc | amplitude | 13 | 20NewsGroup | NA |
| [44] | vqc | amplitude | 10 | ItaCoLa (Italian Corpus of Lin | NA |
| [109] | vqc | amplitude | NR | Kaggle Quantum Programming dat | NA |
| [113] | vqc | amplitude | 9 | Smish (Bangla SMS) | NA |
| [48] | vqc | amplitude | NR | product reviews (dataset name | NA |
| [47] | vqc | amplitude | 10 | CoLA | IBM Kyev backend speci |
| [50] | vqc | amplitude | NR | IFND | NA |
| [52] | vqc | amplitude | NR | 20 Newsgroups (binary subset) | NA |
| [108] | vqc | amplitude | 10 | SST-2 | Amazon Braket quantum |
| [114] | vqc | amplitude | 10 | Custom Aramaic corpus | NA |
| [120] | vqc | amplitude | NR | IMDB | NA |
| [27] | vqc | angle | NR | SARD (Software Assurance Refer | NA |
| [36] | vqc | angle | 10 | UCI ML Drug Review | NA |
| [32] | vqc | angle | 5 | Twitter tweet dataset | IBMQ-manila |
| [84] | vqc | angle | 4 | multilingual Twitter | NA |
| [81] | vqc | angle | NR | SMS spam/ham | NA |
| [85] | vqc | angle | NR | TriviaQA | NA |
| [78] | vqc | angle | NR | Kaggle FIFA World Cup 2022 twe | NA |
| [86] | vqc | angle | 7 | MC | NA |
| [80] | vqc | angle | 16 | IMDb sentence-polarity movie r | NA |
| [79] | vqc | angle | 2 | LIAR | NA |
| [87] | vqc | angle | NR | UMSA-U | NA |
| [83] | vqc | angle | 4 | WordNet | NA |
| [23] | vqc | angle | 4 | Amazon Cell Phones reviews (Ko | simulated noise-free b |
| [82] | vqc | angle | 6 | AI-GA | NA |
| [40] | vqc | angle | 6 | custom Persian poetry humor da | NA |
| [42] | vqc | angle | 3 | Martinican Creole token datase | NA |
| [33] | vqc | angle | NR | 20 Newsgroups | NA |
| [37] | vqc | angle | 4 | Reddit RSDD (Self-Reported Dep | NA |
| [88] | vqc | angle | 5 | MC | NA |
| [143] | vqc | angle | 4 | MC | BAQIS Quafu 156-qubit |
| [70] | vqc | data_reuploading | 6 | ECP-CANDLE P3B3 (cancer pathol | NA |
| [72] | vqc | data_reuploading | 7 | Yelp | IBM Cairo |
| [73] | vqc | data_reuploading | 18 | SST2 | NA |
| [75] | vqc | hybrid_embed | 4 | SATE IV Juliet | NA |
| [74] | vqc | hybrid_embed | 8 | GoEmotions | NA |
| [77] | vqc | other:hamiltonian | 9 | SST | NA |
| [66] | vqc | structured_feature_map | 11 | SelQA | NA |